SMITHIAN FOLD THEORY - INFORMATION SCIENCE BRANCH PAPER 001

From Distinction to Information

An Exact, Parameter-Free and Machine-Closed Derivation of Information Science from Smithian Fold Theory

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Third clean-room reconstruction - complete V1/V2 Information Science reconciliation
Twelve admitted derivations - 11,776 generated candidate structures
Seventy-seven prior obligations closed - none open
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Abstract

This paper reports the completed Information Science branch of the third clean-room reconstruction of Smithian Fold Theory (SFT). From the admitted Foundation and Mathematics receipts it derives symbols and exact distinguishability; encoding and decoding; information quantity; entropy and uncertainty; compression and redundancy; channels and capacity; noise and error; coding; mutual and conditional information; conservation, loss and transformation; classical and probabilistic information; and quantum-information support correspondence. A conventional bit, logarithmic information measure, stochastic cause, fitted distribution, unregistered metric, real or complex amplitude, pretrained model and application result are excluded from derivational premises.

The twelve registered grammars execute 11,776 candidate structures. Every candidate receives an exact decision and each grammar has one all-preserving survivor. Every claim passes minimality, named-shape uniqueness, a depth-independent base/successor certificate, false-premise, source-tamper, artifact-tamper and boundary controls, cryptographic sealing and implementation-distinct recomputation. The branch therefore contains twelve depth-independently closed, model-admitted and independently replicated receipts. Every paper statement is mapped to its registration, complete census, elimination receipt, controls, certificate, source manifest and model-admitted engine receipt. The complete 763-entry V1/V2 source surface has also been reviewed for categorical ownership: 77 atomic Information Science obligations were identified, 77 were reconstructed at equal strength, and none remains open.

The central reconstruction is ledger-valued. Information is complete generated alternative support together with the exact distinctions retained or closed by observation. Entropy retains its observation classes and unresolved pairs rather than collapsing them to an imported logarithmic scalar. Probability is an exact part relation over deterministic support rather than a causal random premise. Channel capacity and error correction follow from exact support separation. Quantum scope is explicitly limited to finite support correspondence; operational quantum laws remain obligations of the later Quantum Computation branch.

Keywords: Smithian Fold Theory; information science; distinguishability; encoding; entropy; uncertainty; compression; channels; coding theory; mutual information; deterministic probability; reversible information; quantum information; open computational science.

1. Central scientific claim

The exact claim of this paper is:

> *Within the frozen SFT V3 Information Science current-knowledge inventory, every one of the twelve registered obligations has a depth-independent, model-admitted and independently replicated engine receipt; all 77 Information Science-owned V1/V2 atomic obligations are closed at equal strength; the inventory contains no unclassified, open or frontier obligation.*

The inventory identity is `sha256:98162a98e2508cb36381ed2b99cb195e3cc7e1b33b8577d4c5a4550a492a0b17`. It fixes scope and dependency order before prose evaluation. Branch closure applies to the exact generated-finite kernels named in the inventory. It does not claim every conventional asymptotic rate theorem, every named code or an operational quantum theory. Fold Protein, Fold Chess, Fold Go, Unison AI and all application experiments were excluded from law selection.

2. Standalone dependency foundation

This version is a same-paper successor patch to Information Science Branch Paper 001 in its existing DOI chain; it does not create a second competing paper. It is standalone in exposition while preserving scientific dependency identities. The Foundation supplies operational occurrence, structural One, complete positive finite count, exact held/whole parts, the minimal Fold, part equivalence, Fold assembly, finite form grammar, canonical form enforcement and the one-way measurement boundary.

The Mathematics branch supplies exact generated arithmetic, finite collections and relations, complete combinatorial families, graphs, algebra, order, finite geometry/topology, deterministic-support probability, optimization, dynamics, proof and composition. These are model-admitted dependencies with their own receipts. They are not silently replaced by conventional axioms. Every dependency path terminates at the sole root theorem, *there is no nothing*, and every Information Science registration has empty axiom and free-parameter tuples.

3. Exact information constitution

Admitted information objects are canonical symbols, held labels, generated words, complete support traces, exact observation classes, unordered source pairs, retained and closed distinction ledgers, exact positive support parts, relations, predecessor fibres, action traces and proof traces. Empty or identity cases use structural empty One. Semantic numerical zero, negative quantity, irrational or imaginary proof values, floating proof arithmetic, completed infinity and an ungenerated continuum remain outside the proof domain.

Host-language integers, Boolean verdicts, empty containers and process status values are implementation mechanics. They count or check artifacts and are not admitted SFT mathematical values. Exact parts use rational held/whole relations; absent cells are empty One rather than a numerical-zero probability. No logarithm is required to force support, uncertainty, dependence, channel separation or correction.

4. What forced, closed and validated mean

Each claim registers a finite product grammar of structural questions before execution. Every axis contains one explicitly rejected coordinate and one all-preserving coordinate. The engine generates their complete Cartesian product in canonical order, hashes every candidate trace, decides every candidate, records every elimination reason and requires exactly one survivor. A familiar name or successful example cannot bypass this census.

Minimality requires that replacing any admitted coordinate loses a registered preservation, completeness, exactness or provenance condition. Named-shape uniqueness requires exactly one all-preserving product coordinate. Generality requires a claim-specific structural-One base and successor certificate, yielding depth-independent closure over generated finite depth without positing a completed infinite object.

Four adverse controls are mandatory: a false premise, changed official source identity, missing/duplicated/additional survivor and excluded boundary import. A distinct Python process then regenerates the literal coordinate domains, candidate order, decision vector, unique survivor, closure flags and controls without importing the scientific law module. Only the shared admission engine may project an independently validated receipt into the model census.

5. Dependency order and executed census

Order	Claim	Candidate structures	Dimensions	Closure
1	`SFT-INFO-SYMBOL-DISTINCTION-001`	512	9	depth-independent
2	`SFT-INFO-ENCODING-DECODING-001`	1,024	10	depth-independent
3	`SFT-INFO-QUANTITY-001`	512	9	depth-independent
4	`SFT-INFO-ENTROPY-UNCERTAINTY-001`	512	9	depth-independent
5	`SFT-INFO-COMPRESSION-REDUNDANCY-001`	1,024	10	depth-independent
6	`SFT-INFO-CHANNEL-CAPACITY-001`	1,024	10	depth-independent
7	`SFT-INFO-NOISE-ERROR-001`	1,024	10	depth-independent
8	`SFT-INFO-CODING-001`	2,048	11	depth-independent
9	`SFT-INFO-MUTUAL-CONDITIONAL-001`	1,024	10	depth-independent
10	`SFT-INFO-CONSERVATION-LOSS-001`	1,024	10	depth-independent

Order	Claim	Candidate structures	Dimensions	Closure
11	SFT-INFO-CLASSICAL-PROBABILISTIC-001	1,024	10	depth-independent
12	SFT-INFO-QUANTUM-CORRESPONDENCE-001	1,024	10	depth-independent

The Information Science total is **11,776** generated candidates and twelve survivors. The complete V3 census through this branch contains 34 admitted derivations and 21,650 generated candidates: 2,450 in Foundation, 7,424 in Mathematics and 11,776 here. Candidate counts describe representation-rule products, not every data object producible by the admitted laws.

6. Derivation 1: Exact symbols, observation classes and distinguishability

Claim identity: `SFT-INFO-SYMBOL-DISTINCTION-001`

6.1 Question, necessity and theorem

Information cannot be quantified before the model derives what a symbol is and exactly when two possible forms remain distinguishable to an observation. Imported alphabets or meanings would select that boundary.

The exact theorem statement is:

> *An admitted symbol is a canonical generated form with a retained held label inside a complete canonical alphabet. Observation is a total single-valued source-bound classification. Two symbols are distinguishable exactly when their retained observation labels differ; symbols sharing one observation image remain distinct microforms but belong to one closed observation class. Every retained and closed distinction is recorded.*

Admitted dependencies: `SFT-FOUNDATION-FORM-ENFORCEMENT-001`, `SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001`, `SFT-MATH-DISCRETE-001`, `SFT-MATH-LOGIC-PROOF-001`.

6.2 Derivation chain, grammar and boundary

Canonical form enforcement supplies exact identity; discrete relations supply total classification; logic supplies distinguished held orientations; the measurement boundary fixes one-way observation. Exhausting nine structural questions forces the symbol/observation-class kernel.

Generation rule: Generate the complete product of alphabet coverage, symbol identity, held label, observation coverage, observation single-valuedness, distinction rule, closed-class retention, finite generality and extra-symbol status.

Exact grammar boundary: All finite symbol systems generated from canonical Fold forms, held labels, complete alphabets, total observation relations and exact retained/closed pair ledgers.

The complete product contains 512 candidates. Its completeness-certificate identity is `sha256:ca1f59aecaef95c9d31d3303d1a4d92673ad4b068e5ef58df58915e37e04becc`. The table states every generating axis, its unfavorable coordinate and its forced coordinate. Their full row-level Cartesian product is preserved in `claims/SFT-INFO-SYMBOL-DISTINCTION-001/candidate_census.json`.

Structural axis	Rejected alternative	Exact rejection	Forced coordinate
`alphabet`	`partial-or-duplicated-alphabet`	Omission or duplication changes the possible symbol family.	`complete-canonical-alphabet` - Every generated symbol form occurs once.
`identity`	`presentation-token`	A presentation token may alias distinct generated forms.	`canonical-form-identity` - The complete construction trace fixes exact symbol identity.
`label`	`unlabelled-form`	Without a retained label the selected fibre identity is unavailable.	`retained-held-label` - The symbol carries its exact held fibre label.
`observation`	`partial-observation`	A missing symbol has no declared observation class.	`total-source-bound-observation` - Every alphabet member has one retained observation row.

Structural axis	Rejected alternative	Exact rejection	Forced coordinate
`image`	`multivalued-image`	Multiple unheld images do not identify the observation made.	`single-retained-image` - Each source symbol has exactly one held observation label.
`distinction`	`assumed-or-visual-difference`	Presentation difference does not establish retained observational separation.	`different-observation-labels` - The observation relation retains distinct images for the two source forms.
`closure`	`source-identity-erased`	Erasing sources prevents an exact account of closed distinctions.	`closed-class-with-microforms` - Merged symbols remain as exact source members of one observation class.
`generality`	`sampled-alphabets`	Examples do not establish extension to a fresh symbol.	`alphabet-successor` - A fresh canonical symbol adds one row and all new distinction pairs.
`addition`	`extra-symbol-semantics`	Imported meaning or frequency is not supplied by form and observation.	`no-extra-symbol-semantics` - Identity and distinction use only generated forms, labels and observation.

6.3 Exhaustion, unique survivor and laws

> *The symbol kernel is a complete canonical held-labelled alphabet, total single-valued observation, exact different-image distinguishability, retained closed classes, successor generality and no imported semantics.*

The engine recorded 512 decisions, 511 eliminations and exactly one survivor, `complete-canonical-alphabet__canonical-form-identity__retained-held-label__total-source-bound-observation__single-retained-image__different-observation-labels__closed-class-with-microforms__alphabet-successor__no-extra-symbol-semantics`. Replacing any survivor coordinate with the rejected coordinate triggers the table's failure condition; the extra-rule coordinate prevents an unforced addition. This proves minimality and named-shape uniqueness inside the frozen grammar. The survivor entails:

- symbol identity is canonical construction identity plus retained held label.
- observation classes partition the complete alphabet without changing microform identity.
- distinguishability is symmetric and irreflexive on exact symbol pairs.
- coarsening closes distinctions only by merging observation images.
- the retained/closed pair ledgers exhaust every generated unordered symbol pair.

6.4 Operational demonstrations and adverse controls

- `total-observation` - The fine observation classifies every symbol exactly once. Result: PASS.
- `fine-distinction` - Fine observation retains every nonidentity pair distinction. Result: PASS.
- `coarse-closure` - Coarse observation closes only the a/b distinction while retaining both microforms. Result: PASS.
- `symmetry` - Swapping a distinguished pair does not change its status. Result: PASS.
- `irreflexivity` - A canonical symbol is not distinguished from itself. Result: PASS.
- `class-retention` - The coarse class retains both exact left microforms. Result: PASS.

The engine-level unfavorable controls are:

- `false_premise` - expected: reject a generated form missing the first required structural condition Observed: Omission or duplication changes the possible symbol family. Result: PASS; receipt `sha256:0fc110df76dc26a6061187a62ca6f4ab63284190dc52235042c8a4baf1aa6440`.
- `tampered_source` - expected: reject a changed official source identity Observed: the changed identity differs from the registered source manifest Result: PASS; receipt `sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26`.
- `tampered_artifact` - expected: reject a missing, duplicated or additional survivor Observed: the generated product contains exactly one registered survivor Result: PASS; receipt `sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219`.

- `boundary` - expected: halt any attempt to import an excluded object or answer-producing model Observed: no imported semantic meaning; no frequency or probability prior; no presentation alias replacing canonical identity; no ungenerated infinite alphabet Result: PASS; receipt `sha256:00999f5bc8b92e3bfaab8395e8e8ddf539ba74e61fe85e7e70092b74d98dcf46`.

6.5 Depth-independent closure

Base: One canonical symbol has one observation row and no nonidentity pair distinction.

Successor: Adding one fresh canonical symbol adds exactly one source-bound observation row and one pair with every prior symbol. Each new pair is classified once as retained or closed by equality of its two observation labels.

The executed product closes the registered representation alternatives. The base/successor certificate extends that rule to every generated finite support size, word depth, source extension or action extension named by the claim. It does not convert finite generation into a completed infinity.

6.6 Meaning and scientific consequence

The first information law makes distinguishability operational rather than intuitive. A symbol is not a typographic mark or imported semantic token; it is a canonical generated form with a held label. An observation is a complete source-bound classification. Two source forms are distinguishable exactly when that classification retains different images. When images coincide, the source forms are not destroyed: they remain the complete microform membership of one closed observation class. This distinction ledger is the substrate from which every later quantity, uncertainty and loss statement is derived.

6.7 Correspondence only after sealing

After the SFT survivor and receipt are fixed, the following established terms may identify translation points: symbol, alphabet, observation, equivalence class, distinguishability. They are not dependencies and did not provide formulas, thresholds or answer tables. A mismatch remains visible as a boundary result rather than being repaired by changing the sealed law.

6.8 Exact exclusions and limitation

- no imported semantic meaning.
- no frequency or probability prior.
- no presentation alias replacing canonical identity.
- no ungenerated infinite alphabet.

This theorem closes finite structural symbols and observation-relative distinction. It does not assign linguistic meaning, occurrence frequency or information quantity; those require downstream laws.

6.9 Machine evidence identities

- Registered status: `independently_replicated`.
- Source manifest: `sha256:9ab9dbeff82654d86ada02cd716d04a4d2a7d5a31358c418f8d5591142804311`.
- Derivation seal: `sha256:78cd2b2e6d03245ba4b96c7c6df45c3b2a2845ab3b1df64b90409384ed195abd`.
- Independent implementation: `sha256:42ac391738828a48baa3b886e4017c8a99a037ff5fe39fce67dd387970cf0f34`.
- Independent certificate: `sha256:75b1bf19a3e78fceca2ea74b37b86f17b74699f90b21d1b5e6b47262f923ae6a`.
- External validation: `sha256:7cec4cb5666884fb6a81239c62e2e2259ee29654bf0143a467aeb8fd4ac85dc1`.
- Engine receipt: `sha256:e387d64c93f5241d90ecf2a447eb6a3518c1e9f473e7ec627d38b7f03ac9d941`.

- Engine receipt path: `receipts/engine/model_admitted/SFT-INFO-SYMBOL-DISTINCTION-001-e387d64c93f5241d.json`.

7. Derivation 2: Exact encoding, decoding and declared representational loss

Claim identity: `SFT-INFO-ENCODING-DECODING-001`

7.1 Question, necessity and theorem

Information requires lawful representation before quantity, compression or channels can be discussed. Encoding must expose exactly which distinctions it preserves and which it closes.

The exact theorem statement is:

> *An admitted encoding is a total single-valued source-bound map from a complete canonical source alphabet into generated codeword support, retaining every source-to-code relation. Exact decoding exists exactly when every used codeword class contains one source and the decoder is the resulting left inverse. A many-to-one encoding is lawful only as declared lossy representation whose complete closed-source classes remain recorded.*

Admitted dependencies: `SFT-FOUNDATION-FOLD-ASSEMBLY-001`, `SFT-MATH-DISCRETE-001`, `SFT-MATH-CATEGORY-TYPE-COMPOSITION-001`, `SFT-INFO-SYMBOL-DISTINCTION-001`.

7.2 Derivation chain, grammar and boundary

Fold assembly supplies generated codewords; symbol law supplies canonical sources and distinctions; discrete maps supply totality; compositional mathematics supplies exact interfaces. Ten structural axes force the total relation, inverse and declared-loss kernel.

Generation rule: Generate the complete product of source coverage, code support, encoding totality, single-valuedness, provenance, decoder law, exact/lossy status, composition, finite generality and extra-code status.

Exact grammar boundary: All finite representations generated from complete canonical source alphabets, generated Fold codewords, total maps, inverse relations, retained source classes and exact interface composition.

The complete product contains 1,024 candidates. Its completeness-certificate identity is `sha256:04b9b004111bd895088946afec50c717e90be78d7193e1d61e5c4c8a4d6c9117`. The table states every generating axis, its unfavorable coordinate and its forced coordinate. Their full row-level Cartesian product is preserved in `claims/SFT-INFO-ENCODING-DECODING-001/candidate_census.json`.

Structural axis	Rejected alternative	Exact rejection	Forced coordinate
`source`	`partial-source`	An omitted source symbol has no representation row.	`complete-canonical-source` - Every source symbol occurs once in the registered domain.
`support`	`free-code-token`	A free token has no generated Fold-word provenance.	`generated-code-support` - Every codeword is an exact generated held-label word.
`coverage`	`partial-map`	A partial map leaves at least one source without a codeword.	`total-source-map` - Every source has one encoding row.
`image`	`unheld-multiple-images`	Multiple images do not identify the codeword actually retained.	`single-retained-codeword` - Each source has exactly one held codeword image.
`provenance`	`source-code-link-erased`	Erasing links prevents reconstruction of source classes.	`complete-source-code-relation` - Every exact source-to-code pair is retained.
`decoder`	`guessed-preimage`	A used codeword can have several source predecessors.	`singleton-class-left-inverse` - Every used code class is singleton and the inverse map returns its source.
`loss`	`silent-loss`	Silent merging falsely claims unavailable distinctions remain.	`declared-closed-source-classes` - Every merged source family is retained as an explicit loss class.
`composition`	`presentation-concatenation`	Concatenation can join unequal code interfaces.	`complete-interface-map` - Every first-stage codeword has exactly one second-stage image.

Structural axis	Rejected alternative	Exact rejection	Forced coordinate
`generality`	`sampled-codebooks`	Examples do not establish a fresh source row.	`source-successor` - A fresh source adds one exact encoding row and updates one code class.
`addition`	`extra-code-convention`	A borrowed numeral or lexical convention can select the code.	`no-extra-code-convention` - Only generated code support and exact relations are used.

7.3 Exhaustion, unique survivor and laws

> *The encoding kernel is complete source and generated code support with a total single-valued provenance relation, singleton-class left-inverse decoding, explicit loss classes, exact composition and no borrowed code.*

The engine recorded 1,024 decisions, 1,023 eliminations and exactly one survivor, `complete-canonical-source__generated-code-support__total-source-map__single-retained-codeword__complete-source-code-relation__singleton-class-left-inverse__declared-closed-source-classes__complete-interface-map__source-successor__no-extra-code-convention`. Replacing any survivor coordinate with the rejected coordinate triggers the table's failure condition; the extra-rule coordinate prevents an unforced addition. This proves minimality and named-shape uniqueness inside the frozen grammar. The survivor entails:

- every source symbol has exactly one retained codeword image.
- lossless exact decoding exists if and only if used codeword classes are singleton.
- many-to-one encoding closes exactly the distinctions inside its retained source classes.
- decoder-after-lossless-encoder returns the original canonical source.
- encoding composition preserves source provenance when every intermediate codeword is mapped.

7.4 Operational demonstrations and adverse controls

- `lossless-total` - The witness lossless relation maps every source once into generated support. Result: PASS.
- `left-inverse` - The exact decoder reconstructs every lossless source. Result: PASS.
- `loss-detected` - The lossy relation has no exact decoder and retains the merged a/b class. Result: PASS.
- `class-ledger` - Every used lossy codeword retains its exact source members. Result: PASS.
- `composition` - Complete code interface composition preserves each source relation. Result: PASS.

The engine-level unfavorable controls are:

- `false_premise` - expected: reject a generated form missing the first required structural condition Observed: An omitted source symbol has no representation row. Result: PASS; receipt `sha256:bf4e85ebd5f2d1856f83756999071a09416cb188f8f2b7575b387df85706d894`.
- `tampered_source` - expected: reject a changed official source identity Observed: the changed identity differs from the registered source manifest Result: PASS; receipt `sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26`.
- `tampered_artifact` - expected: reject a missing, duplicated or additional survivor Observed: the generated product contains exactly one registered survivor Result: PASS; receipt `sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219`.
- `boundary` - expected: halt any attempt to import an excluded object or answer-producing model Observed: no imported binary numeral system; no presentation-only token identity; no silent many-to-one loss; no infinite codebook as a completed object Result: PASS; receipt `sha256:39ff4157377beb066b2a0fa22869a55e623bde5f0bddb7acd4e230983f12d04c`.

7.5 Depth-independent closure

Base: One source symbol maps to one generated codeword and its singleton inverse decodes exactly.

Successor: Adding one fresh source adds exactly one encoding pair. A fresh codeword preserves singleton decoding; a used codeword extends its closed source class and changes the representation status to declared lossy.

The executed product closes the registered representation alternatives. The base/successor certificate extends that rule to every generated finite support size, word depth, source extension or action extension named by the claim. It does not convert finite generation into a completed infinity.

7.6 Meaning and scientific consequence

Encoding is reconstructed as an exact relation, not a convention selected in advance. Every canonical source has one retained generated codeword and every source/code link remains inspectable. Decoding is not guessed: it exists exactly when every used codeword has one source predecessor. A many-to-one representation remains lawful only when it says what it closes and preserves the full merged source class. This establishes lossless and lossy representation before compression, channel or code claims can use those words.

7.7 Correspondence only after sealing

After the SFT survivor and receipt are fixed, the following established terms may identify translation points: alphabet, codeword, encoder, decoder, injective code, lossy representation. They are not dependencies and did not provide formulas, thresholds or answer tables. A mismatch remains visible as a boundary result rather than being repaired by changing the sealed law.

7.8 Exact exclusions and limitation

- no imported binary numeral system.
- no presentation-only token identity.
- no silent many-to-one loss.
- no infinite codebook as a completed object.

This theorem closes finite encoding and decoding structure. It does not yet assign code length value, compression optimality, channel reliability or correction power.

7.9 Machine evidence identities

- Registered status: `independently_replicated`.
- Source manifest: `sha256:d072e49df1f7903aa2987ccb29174b8261c9a32fa26eda6a856c65cee404b0f7`.
- Derivation seal: `sha256:ae856dbb457f937934a2fb5ea450d6902c4460ad478c9c596b8996540c23ebde`.
- Independent implementation: `sha256:d93b45d2158dd50461fcf50bf1ef7ec0668dad57e2a498a1391e4a61fbf7531e`.
- Independent certificate: `sha256:f03b8247debffe52a53757d246405cf4a44e98f22134494df12d6a9e665b2cab`.
- External validation: `sha256:29719e28c425d45cd7a1c62e4a551776a08ff556c85a7bfe87587500c304dbaa`.
- Engine receipt: `sha256:dc3cbc708f5b88b3d9bbae5a646a644dc8feb7fe4a5d6f8367052f5062846d0f`.
- Engine receipt path: `receipts/engine/model_admitted/SFT-INFO-ENCODING-DECODING-001-dc3cbc708f5b88b3.json`.

8. Derivation 3: Exact information quantity as retained distinction structure

Claim identity: `SFT-INFO-QUANTITY-001`

8.1 Question, necessity and theorem

Information quantity must be derived from exact alternatives and observation before entropy, capacity or compression. Importing logarithms would install a conventional numerical answer and forbidden proof values.

The exact theorem statement is:

> *Information quantity is the exact structure of distinguishable alternatives retained from a complete generated support: the support trace, observation-class trace, retained distinction ledger and closed distinction ledger. The first Fold supplies one native distinction unit. Complete Fold-word support across a canonical position trace supplies one independently witnessed unit per position; composition is complete pair support and concatenated position provenance. No logarithm, real scale or semantic zero is foundational.*

Admitted dependencies: `SFT-FOUNDATION-FOLD-ASSEMBLY-001`, `SFT-MATH-EXACT-ARITHMETIC-001`, `SFT-MATH-COMBINATORICS-001`, `SFT-INFO-SYMBOL-DISTINCTION-001`, `SFT-INFO-ENCODING-DECODING-001`.

8.2 Derivation chain, grammar and boundary

Fold assembly supplies complete word support; symbol distinction supplies exact observation ledgers; encoding supplies representation; combinatorics supplies complete pairs. Nine structural questions force a ledger-valued quantity and independently witnessed Fold-position unit.

Generation rule: Generate the complete product of support coverage, distinction ledger, observation classes, native unit, position independence, quantity representation, composition, generality and extra-scale status.

Exact grammar boundary: All finite information profiles generated from complete canonical Fold-word support, total observation classes, exact pair distinctions, independently varied positions and pair-cell composition.

The complete product contains 512 candidates. Its completeness-certificate identity is `sha256:40d0d2569464d104f73e01fed5c7bb9ef0cb4640c1d24709f6877e3674152e3c`. The table states every generating axis, its unfavorable coordinate and its forced coordinate. Their full row-level Cartesian product is preserved in `claims/SFT-INFO-QUANTITY-001/candidate_census.json`.

Structural axis	Rejected alternative	Exact rejection	Forced coordinate
`support`	`partial-support`	Omitting a generated alternative changes every distinction count.	`complete-canonical-support` - Every registered alternative occurs once.
`pairs`	`sampled-pairs`	Sampling cannot prove a distinction ledger complete.	`complete-unordered-pair-ledger` - Every pair of distinct support forms occurs once.
`observation`	`assumed-resolution`	An assumed resolution has no source-to-image evidence.	`exact-observation-classes` - Image equality classifies every pair as retained or closed.
`unit`	`imported-bit`	A conventional binary unit would select the answer.	`first-Fold-distinction` - The minimal Fold's two held labels supply the first exact alternative distinction.
`positions`	`word-length-only`	Length alone does not show each position can vary independently.	`independent-position-witness` - Every position realizes the Fold distinction with all other positions held.
`quantity`	`real-logarithm`	A logarithm imports irrational and floating proof values.	`exact-support-and-distinction-profile` - The complete exact ledgers retain the quantity without a borrowed scale.
`composition`	`count-product-only`	A scalar product erases source-coordinate provenance.	`complete-pair-support` - Every left state pairs once with every right state and retains both word traces.
`generality`	`depth-table`	A table of small supports does not prove the next position.	`position-successor` - A fresh position appends every held label to every prior word.
`addition`	`extra-log-base-or-unit`	A chosen base or physical scale is a free parameter.	`no-extra-information-scale` - All quantity follows from Fold units, support and observation.

8.3 Exhaustion, unique survivor and laws

> *The information-quantity kernel is complete support and pair ledgers, exact retained/closed observation distinctions, first-Fold units, independently witnessed position traces, pair-support composition and no log scale.*

The engine recorded 512 decisions, 511 eliminations and exactly one survivor, `complete-canonical-support__complete-unordered-pair-ledger__exact-observation-classes__first-Fold-distinction__independent-position-witness__exact-support-and-distinction-profile__complete-pair-support__position-successor__no-extra-information-scale`. Replacing any survivor coordinate with the rejected coordinate triggers the table's failure condition; the extra-rule coordinate prevents an unforced addition. This proves minimality and named-shape uniqueness inside the frozen grammar. The survivor entails:

- retained and closed distinction ledgers are disjoint and exhaust complete support pairs.
- fine observation retains every nonidentity support distinction.
- coarsening can only move pair distinctions from retained to closed.
- one complete Fold position supplies one independently realized distinction unit.
- composing supports retains both source coordinates and concatenates their unit traces.

8.4 Operational demonstrations and adverse controls

- `complete-support` - Three independently held Fold positions generate eight unique exact words. Result: PASS.
- `fine-ledger` - Fine observation retains every distinct support pair and closes none. Result: PASS.
- `coarse-ledger` - First-position observation retains cross-class pairs and closes within-class pairs. Result: PASS.
- `position-units` - Each of three positions independently realizes the Fold distinction. Result: PASS.
- `support-composition` - Two two-word supports compose into four coordinate-preserving words. Result: PASS.

The engine-level unfavorable controls are:

- `false_premise` - expected: reject a generated form missing the first required structural condition Observed: Omitting a generated alternative changes every distinction count. Result: PASS; receipt `sha256:66c8c2acb6c7e2805130d894dcbdcab51bd9665d1d065a1ca9860f3b23d14826`.
- `tampered_source` - expected: reject a changed official source identity Observed: the changed identity differs from the registered source manifest Result: PASS; receipt `sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26`.
- `tampered_artifact` - expected: reject a missing, duplicated or additional survivor Observed: the generated product contains exactly one registered survivor Result: PASS; receipt `sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219`.
- `boundary` - expected: halt any attempt to import an excluded object or answer-producing model Observed: no logarithm as a proof operation; no real-valued or irrational information quantity; no semantic numerical zero for identity information; no incomplete support presented as full quantity Result: PASS; receipt `sha256:40cc647fb3b88cb6aca2ec0f86b8a4bc0e2a9c7e1810685d2ca3f5ce2df95355`.

8.5 Depth-independent closure

Base: The first Fold position has complete two-label support and one independently realized distinction unit.

Successor: Appending one fresh position extends every prior word once with every held label. Prior position witnesses remain, the fresh position varies across its labels for each held prefix, and all new support pairs enter exactly once.

The executed product closes the registered representation alternatives. The base/successor certificate extends that rule to every generated finite support size, word depth, source extension or action extension named by the claim. It does not convert finite generation into a completed infinity.

8.6 Meaning and scientific consequence

Information quantity is carried by exact alternatives and their distinguishability, not by a borrowed real scale. The kernel retains complete word support, every unordered source pair, every observation class and the partition of pairs into retained and closed distinctions. The first Fold contributes the first exact two-label distinction. A longer word contributes another native unit only when that position can vary independently while the remaining positions are held. Logarithmic units may summarize a sealed finite support in correspondence, but they are not used to force the support or its distinctions.

8.7 Correspondence only after sealing

After the SFT survivor and receipt are fixed, the following established terms may identify translation points: information content, bit, alphabet size, logarithmic information, joint information. They are not dependencies and did not provide formulas, thresholds or answer tables. A mismatch remains visible as a boundary result rather than being repaired by changing the sealed law.

8.8 Exact exclusions and limitation

- no logarithm as a proof operation.
- no real-valued or irrational information quantity.
- no semantic numerical zero for identity information.
- no incomplete support presented as full quantity.

This theorem closes exact finite information quantity. Conventional logarithmic units may be post-seal correspondence summaries but are not SFT proof values.

8.9 Machine evidence identities

- Registered status: `independently_replicated`.
- Source manifest: `sha256:6a918de5f716d1371af1e475ce3a36d565c12e6d92f16ef430315c75dc2dad93`.
- Derivation seal: `sha256:268633b173cd9292188e15669163580ed1e76b99b66c861dd70181087b23bb43`.
- Independent implementation: `sha256:67a80bd1df98d114b370eecae171e57f012f3a5c86e3056fb2f24c2f9161e479`.
- Independent certificate: `sha256:5dddce624d62ebceb275a5a87beda0fe6886bcd348a59919c5145ac5af4b56ac`.
- External validation: `sha256:325c8dcdbfc5b20ce21c7549baac1fe63ca0eabb138178d09b8f9b7eef013b9b`.
- Engine receipt: `sha256:a06381112b197ea1762d8dd06197c87f03055a4554429f100ead4a93bbb2362d`.
- Engine receipt path: `receipts/engine/model_admitted/SFT-INFO-QUANTITY-001-a06381112b197ea1.json`.

9. Derivation 4: Exact entropy and uncertainty from deterministic observation classes

Claim identity: `SFT-INFO-ENTROPY-UNCERTAINTY-001`

9.1 Question, necessity and theorem

Entropy and uncertainty must be reconstructed without treating randomness or real logarithms as premises. Exact observation classes already contain the complete unresolved information.

The exact theorem statement is:

> *Uncertainty is the complete ledger of deterministic microstate distinctions closed inside each exact observation class. Entropy is the provenance-retaining family of those classes, their exact positive support parts, microstate traces and unresolved-pair ledgers; singleton certainty uses structural empty One. Refining observation can only remove closed pairs, while coarsening can only add them. No stochastic cause, logarithm, numerical zero or floating expectation enters the law.*

Admitted dependencies: `SFT-FOUNDATION-PART-EQUIVALENCE-001`, `SFT-MATH-PROBABILITY-STATISTICS-001`, `SFT-INFO-SYMBOL-DISTINCTION-001`, `SFT-INFO-QUANTITY-001`.

9.2 Derivation chain, grammar and boundary

Mathematical probability supplies exact support parts; symbol distinction supplies observation classes; information quantity supplies retained/closed ledgers. Nine axes force an entropy object that preserves every class, part and unavailable distinction.

Generation rule: Generate the complete product of microstate support, observation partition, class weights, unresolved pairs, entropy representation, refinement order, deterministic status, generality and extra-entropy status.

Exact grammar boundary: All finite uncertainty structures generated from complete deterministic microstate support, total observation partitions, exact class parts, complete within-class pair ledgers and refinement relations.

The complete product contains 512 candidates. Its completeness-certificate identity is `sha256:77baae24e3e22695c0246fc61c48d8b34f73542f2bd557e1407861f24f21339d`. The table states every generating axis, its unfavorable coordinate and its forced coordinate. Their full row-level Cartesian product is preserved in `claims/SFT-INFO-ENTROPY-UNCERTAINTY-001/candidate_census.json`.

Structural axis	Rejected alternative	Exact rejection	Forced coordinate
`support`	`partial-microstate-support`	Omission changes observation classes and uncertainty.	`complete-deterministic-support` - Every registered microstate occurs once.
`partition`	`overlapping-or-partial-classes`	Overlap or omission does not partition exact support.	`total-disjoint-observation-classes` - Every microstate belongs to exactly one observed class.
`weights`	`floating-probability`	A floating propensity imports precision and stochastic cause.	`exact-class-whole-part` - The class count is retained relative to complete support count.
`unresolved`	`scalar-uncertainty-score`	A scalar score erases which distinctions are unavailable.	`complete-within-class-pairs` - Every pair of distinct class microstates is retained once.
`entropy`	`logarithmic-expectation`	Logarithms and real expectations import forbidden proof values.	`weighted-class-distinction-ledger` - Every class retains exact part, members and unresolved distinctions.
`refinement`	`assumed-more-information`	A name does not prove one observation separates every fine class.	`class-containment-witness` - Every fine class lies inside one coarse class and its closed-pair set is contained.
`cause`	`ontic-random-source`	A stochastic source is neither generated nor required.	`deterministic-support-observation` - Uncertainty comes from distinctions closed by observation over deterministic support.
`generality`	`sampled-distributions`	Examples cannot establish a fresh microstate class update.	`microstate-successor` - A fresh state extends exactly one class and its new within-class pairs.
`addition`	`extra-log-base-or-prior`	A chosen log base or prior is a free parameter.	`no-extra-entropy-convention` - The exact ledger follows only from support and observation.

9.3 Exhaustion, unique survivor and laws

> *The entropy/uncertainty kernel is complete deterministic support, total observation classes, exact class parts, complete unresolved-pair ledgers, refinement containment, successor closure and no stochastic or log rule.*

The engine recorded 512 decisions, 511 eliminations and exactly one survivor, `complete-deterministic-support__total-disjoint-observation-classes__exact-class-whole-part__complete-within-class-pairs__weighted-class-distinction-ledger__class-containment-witness__deterministic-support-observation__microstate-successor__no-extra-entropy-convention`. Replacing any survivor coordinate with the rejected coordinate triggers the table's failure condition; the extra-rule coordinate prevents an unforced addition. This proves minimality and named-shape uniqueness inside the frozen grammar. The survivor entails:

- observation classes form a disjoint complete partition of deterministic support.
- uncertainty is exactly the set of within-class nonidentity microstate pairs.
- singleton classes carry empty-One uncertainty rather than numerical zero.
- fine observation cannot have more closed pairs than a coarser observation it refines.
- entropy retains class provenance and exact part weights instead of collapsing to a floating scalar.

9.4 Operational demonstrations and adverse controls

- `fine-certainty` - Fine observation gives every state a singleton class with empty-One uncertainty. Result: PASS.
- `prefix-uncertainty` - Prefix observation has two exact half-support classes and one unresolved pair each. Result: PASS.
- `coarse-uncertainty` - Complete coarsening retains all six microstate pairs as unresolved. Result: PASS.
- `refinement-order` - Fine refines prefix, and prefix refines the one-class observation. Result: PASS.
- `closed-pair-monotonicity` - Refinement reduces closed distinctions exactly in the witness chain. Result: PASS.

The engine-level unfavorable controls are:

- `false_premise` - expected: reject a generated form missing the first required structural condition Observed: Omission changes observation classes and uncertainty. Result: PASS; receipt `sha256:b8cc7b2826b60bb835dce41750f0f9ed5e6ec9175437b2a571fbdb1981eea15a`.
- `tampered_source` - expected: reject a changed official source identity Observed: the changed identity differs from the registered source manifest Result: PASS; receipt `sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26`.
- `tampered_artifact` - expected: reject a missing, duplicated or additional survivor Observed: the generated product contains exactly one registered survivor Result: PASS; receipt `sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219`.
- `boundary` - expected: halt any attempt to import an excluded object or answer-producing model Observed: no stochastic source premise; no logarithmic entropy proof value; no numerical zero for singleton uncertainty; no fitted probability distribution Result: PASS; receipt `sha256:e2deb036c0db8c8405ccf1fc3515c5acfa97882e8e851aed41155d83d8769561`.

9.5 Depth-independent closure

Base: One deterministic microstate forms one exact whole class with structural empty-One uncertainty.

Successor: Adding one fresh microstate assigns it to one retained observation image, updates that class's exact part and adds exactly one unresolved pair with every prior member of the same class; other classes remain unchanged.

The executed product closes the registered representation alternatives. The base/successor certificate extends that rule to every generated finite support size, word depth, source extension or action extension named by the claim. It does not convert finite generation into a completed infinity.

9.6 Meaning and scientific consequence

Uncertainty is the exact set of deterministic microstate pairs an observation cannot separate. Entropy is the provenance-retaining family of observation classes, exact positive support parts, class members and unresolved pair ledgers. A singleton class has structural empty-One uncertainty. Refining observation removes closed pairs; coarsening adds them. Nothing in this construction requires an ontic random cause, a logarithm, an expectation over reals or a fitted distribution. The result preserves more evidence than a scalar because every unresolved distinction remains named.

9.7 Correspondence only after sealing

After the SFT survivor and receipt are fixed, the following established terms may identify translation points: entropy, uncertainty, partition entropy, resolution, deterministic ensemble. They are not dependencies and did not provide formulas, thresholds or answer tables. A mismatch remains visible as a boundary result rather than being repaired by changing the sealed law.

9.8 Exact exclusions and limitation

- no stochastic source premise.
- no logarithmic entropy proof value.
- no numerical zero for singleton uncertainty.
- no fitted probability distribution.

This theorem closes exact finite entropy and uncertainty. Conventional scalar Shannon entropies are post-seal summaries only and require a separately declared correspondence map.

9.9 Machine evidence identities

- Registered status: `independently_replicated`.
- Source manifest: `sha256:7154a737d0cc6c3db65e1dc129773dc38ede23ec9a78c5db34da4bbebedd72e1`.
- Derivation seal: `sha256:d7ea4b7d51ccdbda737ac0ac8118ef0fa25f6d93544a0e26bcdc3284fd713682`.
- Independent implementation: `sha256:b608e2eb5217eb81c46115f9952ae59f5d2c1366842cb508f2af62b8b9d856c1`.
- Independent certificate: `sha256:b2f70d4486625da56092d8d2bcee3691720ffc312a91197f366adbd3d30520a9`.
- External validation: `sha256:2db327cc0f0f7585ef9ad15422f5f94e4d17686e69312321ab272b5754be99ae`.
- Engine receipt: `sha256:d604932214499ea1e11fc2df7b71f4ea0b67014e3001023720b325d6203d2b72`.
- Engine receipt path: `receipts/engine/model_admitted/SFT-INFO-ENTROPY-UNCERTAINTY-001-d604932214499ea1.json`.

10. Derivation 5: Exact lossless compression and redundancy accounting

Claim identity: `SFT-INFO-COMPRESSION-REDUNDANCY-001`

10.1 Question, necessity and theorem

Compression claims can hide information in a dictionary, parser, model or decoder. The law must force exact reconstruction and count every retained resource before shortening has scientific meaning.

The exact theorem statement is:

> *Lossless compression is a complete description relation whose retained dictionary and token trace reconstruct the exact canonical message. Redundancy is a message distinction or occurrence reconstructible from retained description structure. Variable codewords are uniquely parseable only with a complete delimiter or prefix-free witness. Every dictionary and token resource is counted; no fixed complete word support can injectively map all its forms into the strictly shorter nonempty support.*

Admitted dependencies: `SFT-MATH-COMBINATORICS-001`, `SFT-MATH-OPTIMIZATION-001`, `SFT-INFO-ENCODING-DECODING-001`, `SFT-INFO-QUANTITY-001`.

10.2 Derivation chain, grammar and boundary

Encoding supplies exact decoders; information quantity supplies resource traces; combinatorics supplies full fixed and shorter supports; optimization retains all minimal descriptions. Ten axes force lossless reconstruction, complete accounting and the universal-shortening boundary.

Generation rule: Generate the complete product of message coverage, description provenance, exact reconstruction, parsing, redundancy witness, resource accounting, shortening status, optimization, generality and extra-compressor status.

Exact grammar boundary: All finite lossless descriptions generated from canonical messages, retained dictionaries, token traces, exact decoders, unique parsing witnesses and complete positive resource traces.

The complete product contains 1,024 candidates. Its completeness-certificate identity is `sha256:428cb09aa9b7cb3be4e50612a4d3757c46b45e5fae06603e4780eb4ea7cd847a`. The table states every generating axis, its unfavorable coordinate and its forced coordinate. Their full row-level Cartesian product is preserved in `claims/SFT-INFO-COMPRESSION-REDUNDANCY-001/candidate_census.json`.

Structural axis	Rejected alternative	Exact rejection	Forced coordinate
`message`	`partial-message`	An omitted occurrence changes the reconstruction target.	`complete-canonical-message` - Every source occurrence remains in exact order.
`description`	`opaque-short-string`	A short string without dictionary provenance cannot reconstruct the source.	`dictionary-and-token-trace` - Every reusable block and held token occurrence is retained.
`reconstruction`	`similar-output`	Similarity does not establish exact source identity.	`exact-decoder-equality` - Decoding reproduces the complete canonical message trace.
`parsing`	`ambiguous-concatenation`	One stream may admit several codeword parses.	`prefix-or-delimiter-witness` - A generated boundary makes the parse unique.
`redundancy`	`frequency-assumption`	Frequency alone does not prove reconstruction.	`reconstructible-occurrence-trace` - Omitted raw occurrences are generated exactly from retained description structure.
`resources`	`payload-only`	Ignoring a dictionary or decoder hides required information.	`complete-description-resource` - Dictionary, tokens, boundaries and decoder identity are all retained.
`shortening`	`universal-shortening-claim`	Strictly shorter support has fewer generated nonempty codewords than complete fixed support.	`pigeonhole-boundary` - At least one full-support message cannot receive a distinct strictly shorter codeword.
`optimization`	`tuned-score`	A floating objective or tie rule is unforced.	`complete-undominated-descriptions` - All exact descriptions are generated and every minimal resource trace tie is retained.
`generality`	`sampled-messages`	Examples do not establish a fresh repeated token.	`token-successor` - Appending one retained token reconstructs exactly its registered dictionary block.
`addition`	`extra-statistical-model`	A learned or fitted model can select the representation.	`no-extra-compression-model` - Only exact generated dictionaries, tokens and decoding are used.

10.3 Exhaustion, unique survivor and laws

> *The compression kernel is exact message reconstruction from a uniquely parseable dictionary/token description, complete resource accounting, reconstructible redundancy, pigeonhole-limited shortening and no fitted model.*

The engine recorded 1,024 decisions, 1,023 eliminations and exactly one survivor, `complete-canonical-message__dictionary-and-token-trace__exact-decoder-equality__prefix-or-delimiter-witness__reconstructible-occurrence-trace__complete-description-resource__pigeonhole-boundary__complete-undominated-descriptions__token-successor__no-extra-compression-model`. Replacing any survivor coordinate with the rejected coordinate triggers the table's failure condition; the extra-rule coordinate prevents an unforced addition. This proves minimality and named-shape uniqueness inside the frozen grammar. The survivor entails:

- losslessness is exact equality of reconstructed and source canonical traces.
- a reusable block is redundancy only when every replaced occurrence is reconstructible.
- dictionary, parser and decoder resources belong to the complete description.
- prefix freedom or an explicit delimiter is required for unique variable-word parsing.
- no injective code can assign every complete fixed-width message a strictly shorter nonempty word.

10.4 Operational demonstrations and adverse controls

- `exact-reconstruction` - Three retained block tokens reconstruct the six-cell message exactly. Result: PASS.
- `resource-accounting` - The description retains one dictionary entry and all three token occurrences. Result: PASS.
- `redundancy-provenance` - Every reconstructed raw cell retains the token and dictionary block that supplied it. Result: PASS.
- `prefix-control` - The witness prefix family parses uniquely while a prefix-containing family fails. Result: PASS.
- `universal-bound` - Complete two-label width-three support cannot inject into all shorter nonempty words. Result: PASS.

The engine-level unfavorable controls are:

- `false_premise` - expected: reject a generated form missing the first required structural condition Observed: An omitted occurrence changes the reconstruction target. Result: PASS; receipt `sha256:ba3756b2fcba103d3088afc0d25c904307b3e90947e5ff6158b0a2adf6b9c490`.
- `tampered_source` - expected: reject a changed official source identity Observed: the changed identity differs from the registered source manifest Result: PASS; receipt `sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26`.
- `tampered_artifact` - expected: reject a missing, duplicated or additional survivor Observed: the generated product contains exactly one registered survivor Result: PASS; receipt `sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219`.
- `boundary` - expected: halt any attempt to import an excluded object or answer-producing model Observed: no hidden learned model or trained weights; no payload-only size claim; no ambiguous variable-word parsing; no universal lossless shortening claim Result: PASS; receipt `sha256:ca800897b20ecc2e2616fa2a070b9de583c2e1943f29ea17d3036b4837bc734e`.

10.5 Depth-independent closure

Base: One dictionary token reconstructs its one registered nonempty block and retains the complete description.

Successor: Appending one valid held token appends exactly its dictionary block to the reconstructed message and one token cell to the resource trace. Exact reconstruction and parsing remain local and replayable.

The executed product closes the registered representation alternatives. The base/successor certificate extends that rule to every generated finite support size, word depth, source extension or action extension named by the claim. It does not convert finite generation into a completed infinity.

10.6 Meaning and scientific consequence

Compression is lawful only when the retained dictionary, token trace, parser and decoder reconstruct the exact source. Resource comparison therefore counts the entire description system rather than reporting a shortened payload while hiding its dictionary. Redundancy is a structurally identified repeated or dependent position, not padding inferred from frequency. The complete-support control proves the finite pigeonhole boundary: a one-to-one code cannot assign every word a strictly shorter nonempty generated word. Compression gains are consequently source-family and representation dependent, never an unqualified universal shortening law.

10.7 Correspondence only after sealing

After the SFT survivor and receipt are fixed, the following established terms may identify translation points: lossless compression, redundancy, prefix code, dictionary coding, pigeonhole bound. They are not dependencies and did not provide formulas, thresholds or answer tables. A mismatch remains visible as a boundary result rather than being repaired by changing the sealed law.

10.8 Exact exclusions and limitation

- no hidden learned model or trained weights.
- no payload-only size claim.
- no ambiguous variable-word parsing.
- no universal lossless shortening claim.

This theorem closes finite structural compression. It does not assert a universal best compressor, a probability distribution or an asymptotic real-valued rate.

10.9 Machine evidence identities

- Registered status: `independently_replicated`.

- Source manifest: `sha256:72926e8f640dbad6cd14100e6e87e111f7b31aa9df24914e176632f7f35642fc`.
- Derivation seal: `sha256:075208fb299525387beb95e18f7ad83f1b6f26c24e32f0789b59188928e88bbe`.
- Independent implementation: `sha256:483dc4a3279cd2a7deaa5358f916e771d8ca79dbfa9d75800d974f5c33891315`.
- Independent certificate: `sha256:69db6a7db7c8c5c8edcf2ae986c4fc900de7b382cc89b190e73f5639e9072cda`.
- External validation: `sha256:0ca99dbf97b53f602285051ef02a3449aee5b56396b921c8e1cd87d76da9f10f`.
- Engine receipt: `sha256:4d46ed451a5cd3c3a19fcb9d8b8cc3c32ffc8f92a9cc1d55cfb93f331023ee07`.
- Engine receipt path: `receipts/engine/model_admitted/SFT-INFO-COMPRESSION-REDUNDANCY-001-4d46ed451a5cd3c3.json`.

11. Derivation 6: Exact finite channels, confusability and capacity

Claim identity: `SFT-INFO-CHANNEL-CAPACITY-001`

11.1 Question, necessity and theorem

Capacity must follow from which source distinctions a channel can preserve, not from a borrowed probabilistic rate formula. Complete finite transport and confusability make that boundary exact.

The exact theorem statement is:

> *An admitted channel is a complete canonical input carrier, complete output carrier and held source-bound transport relation giving every input at least one output. Inputs are confusable exactly when their output supports overlap. A zero-error code is a held input selection with pairwise disjoint output supports. Exact channel capacity is the complete family of greatest-cardinality distinguishable codes, with all ties retained; composition is relational path composition at an equal interface.*

Admitted dependencies: `SFT-MATH-GRAPH-NETWORK-001`, `SFT-MATH-COMBINATORICS-001`, `SFT-MATH-OPTIMIZATION-001`, `SFT-INFO-ENCODING-DECODING-001`, `SFT-INFO-QUANTITY-001`.

11.2 Derivation chain, grammar and boundary

Graphs supply relations and paths; encoding supplies messages and codewords; combinatorics exhausts selections; optimization retains all greatest codes; information quantity supplies exact distinction meaning. Ten axes force the finite zero-error channel and capacity family.

Generation rule: Generate the complete product of input coverage, output coverage, relation provenance, transport totality, confusability, code generation, capacity selection, composition, generality and extra-channel status.

Exact grammar boundary: All finite channels generated from canonical carriers, held transport relations, complete output supports, pairwise confusability checks, exhaustive input selections and exact relation composition.

The complete product contains 1,024 candidates. Its completeness-certificate identity is `sha256:0f4bdf024d8eab5ad6d448ceb713d2eddad5e4fcac806ddb933fabebf43d7917`. The table states every generating axis, its unfavorable coordinate and its forced coordinate. Their full row-level Cartesian product is preserved in `claims/SFT-INFO-CHANNEL-CAPACITY-001/candidate_census.json`.

Structural axis	Rejected alternative	Exact rejection	Forced coordinate
`inputs`	`partial-input-carrier`	An omitted input changes possible messages and capacity.	`complete-canonical-inputs` - Every registered input occurs once.
`outputs`	`partial-output-carrier`	An omitted output hides a possible transport image.	`complete-canonical-outputs` - Every registered output occurs once.

Structural axis	Rejected alternative	Exact rejection	Forced coordinate
relation	opaque-channel-box	An opaque box hides source-to-output alternatives.	held-transport-relation - Every possible source/output pair is retained.
totality	missing-input-row	An input without an output is outside the claimed channel action.	nonempty-output-support-per-input - Every input has at least one retained image.
confusability	probability-threshold	A threshold imports a distribution and free scale.	overlapping-output-support - Confusability is exact shared output membership.
codes	sampled-input-set	Sampling cannot establish every pair separated.	pairwise-disjoint-output-selection - Every selected input pair has disjoint output support.
capacity	logarithmic-rate	A real logarithmic rate imports scale and asymptotic assumptions.	complete-greatest-code-family - Every input selection is generated and all maximum distinguishable ties survive.
composition	endpoint-guess	Endpoints alone omit the shared interface path.	exact-relational-paths - Every source-to-middle and middle-to-target path generates one composite cell.
generality	sampled-channels	Examples do not establish a fresh input relation.	carrier-successor - A fresh input adds its complete output support and all new confusability pairs.
addition	extra-transition-probability	A probability law is not supplied by the transport relation.	no-extra-channel-model - Capacity follows only from exact relation and distinguishability.

11.3 Exhaustion, unique survivor and laws

> *The channel kernel is complete input/output carriers and held total transport, exact overlap confusability, exhaustive zero-error codes, the complete greatest code family, relational composition and no stochastic rule.*

The engine recorded 1,024 decisions, 1,023 eliminations and exactly one survivor, `complete-canonical-inputs__complete-canonical-outputs__held-transport-relation__nonempty-output-support-per-input__overlapping-output-support__pairwise-disjoint-output-selection__complete-greatest-code-family__exact-relational-paths__carrier-successor__no-extra-channel-model`. Replacing any survivor coordinate with the rejected coordinate triggers the table's failure condition; the extra-rule coordinate prevents an unforced addition. This proves minimality and named-shape uniqueness inside the frozen grammar. The survivor entails:

- channel output support retains every possible output of one exact input.
- confusability is symmetric overlap of output supports.
- a zero-error input code has pairwise disjoint output supports.
- capacity retains every greatest distinguishable selection rather than applying an undeclared tie rule.
- composed output paths cannot restore an input distinction already merged at an intermediate interface.

11.4 Operational demonstrations and adverse controls

- `clean-capacity` - Disjoint clean outputs retain all three inputs as one greatest code. Result: PASS.
- `merged-confusability` - Inputs a and b are confusable exactly through shared output x. Result: PASS.
- `tied-capacity` - The merged channel retains both greatest two-input alternatives rather than tie-breaking. Result: PASS.
- `code-control` - The a/b selection fails zero-error distinction while a/c passes. Result: PASS.
- `composition` - Composing two clean identity-interface channels preserves exact source/output paths. Result: PASS.

The engine-level unfavorable controls are:

- `false_premise` - expected: reject a generated form missing the first required structural condition Observed: An omitted input changes possible messages and capacity. Result: PASS; receipt `sha256:d2e6c5c6e7916f77a1634ae43952a1a0fd1cee72f93edf62ebee79333ab10d18`.

- `tampered_source` - expected: reject a changed official source identity Observed: the changed identity differs from the registered source manifest Result: PASS; receipt `sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26`.
- `tampered_artifact` - expected: reject a missing, duplicated or additional survivor Observed: the generated product contains exactly one registered survivor Result: PASS; receipt `sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219`.
- `boundary` - expected: halt any attempt to import an excluded object or answer-producing model Observed: no transition probability or stochastic channel cause; no floating error threshold; no logarithmic or asymptotic capacity proof value; no ungenerated infinite code family Result: PASS; receipt `sha256:213dbc3640d5133d828b84453a6831298f15d164097b8958314613d00e7e33fc`.

11.5 Depth-independent closure

Base: One input with one nonempty output support forms a one-member distinguishable code family.

Successor: Adding one fresh input adds its exact output support and one confusability decision with every prior input. Every prior code either extends by the fresh input when all supports are disjoint or remains unchanged.

The executed product closes the registered representation alternatives. The base/successor certificate extends that rule to every generated finite support size, word depth, source extension or action extension named by the claim. It does not convert finite generation into a completed infinity.

11.6 Meaning and scientific consequence

A channel is a complete relation from each input to every output reachable through a registered path. Inputs are confusable exactly when their output supports overlap. A zero-error code is a held input family with pairwise disjoint output supports. Capacity is not imported as a logarithm or fitted rate: it is the complete family of greatest distinguishable input selections, with ties retained. Composition joins channels only across an exact interface and preserves path provenance, making every limitation traceable to concrete overlaps.

11.7 Correspondence only after sealing

After the SFT survivor and receipt are fixed, the following established terms may identify translation points: communication channel, confusability graph, zero-error code, channel capacity, channel composition. They are not dependencies and did not provide formulas, thresholds or answer tables. A mismatch remains visible as a boundary result rather than being repaired by changing the sealed law.

11.8 Exact exclusions and limitation

- no transition probability or stochastic channel cause.
- no floating error threshold.
- no logarithmic or asymptotic capacity proof value.
- no ungenerated infinite code family.

This theorem closes finite zero-error structural capacity. Probabilistic error rates and asymptotic coding theorems require separately generated finite evidence and cannot be imported here.

11.9 Machine evidence identities

- Registered status: `independently_replicated`.
- Source manifest: `sha256:19a1d66b7f0222f90955bb27db339eb93e576cd935d33e9838d6b62b2a9acd00`.
- Derivation seal: `sha256:c98482b0294a53ba5676104d268e34c95322175800e0d6dd54553aefe8f9d37c`.
- Independent implementation: `sha256:a14b1fa8e95876ffaf760fdca476e7870a63ed4b7135aa4e37e6fb33a845da75`.

- Independent certificate: `sha256:c036dd4a381bdeb6aeb7d0802f54991db22bfa0febf3fd47ac834369923a017c`.
- External validation: `sha256:fd90d9edab2dff3e7ed79b658fc48d9c66e285fe265229537de02f86718c89b5`.
- Engine receipt: `sha256:fb47c7f7be3e832cb67658042b2c7c3224113d9262cb4c45df7924a1f3a1cc01`.
- Engine receipt path: `receipts/engine/model_admitted/SFT-INFO-CHANNEL-CAPACITY-001-fb47c7f7be3e832c.json`.

12. Derivation 7: Exact noise, error, detection and correction boundaries

Claim identity: `SFT-INFO-NOISE-ERROR-001`

12.1 Question, necessity and theorem

Coding cannot be derived until the model states exactly what noise changes, what detection observes and why a correction is unique. Probabilities and metrics are not necessary for these structural boundaries.

The exact theorem statement is:

> *Noise is a registered source-to-received transformation relation over complete canonical word carriers; error is the exact changed-position trace between a held source and received form. Detection is forced when a nonidentity image lies outside valid code support. Correction is forced only when the received form has one registered source predecessor; multiple predecessors remain an explicit closed source class. No random cause, error probability, distance threshold or unrecorded predecessor inference is admitted.*

Admitted dependencies: `SFT-MATH-GRAPH-NETWORK-001`, `SFT-MATH-DYNAMICAL-SYSTEMS-001`, `SFT-INFO-CHANNEL-CAPACITY-001`, `SFT-INFO-ENTROPY-UNCERTAINTY-001`.

12.2 Derivation chain, grammar and boundary

Channels supply source/output relations; dynamics supplies transitions and predecessor loss; entropy supplies closed classes; graphs supply paths. Ten structural axes force exact action provenance, error traces and singleton-predecessor correction.

Generation rule: Generate the complete product of source support, received support, transformation provenance, error trace, detection, predecessor classes, correction, deterministic status, generality and extra-noise status.

Exact grammar boundary: All finite error systems generated from canonical source and received words, held transformation actions, exact changed-position traces, valid code support and complete predecessor classes.

The complete product contains 1,024 candidates. Its completeness-certificate identity is `sha256:65da8756ca47477887e29bed39c882022116b985ee68ad461d8e24e26a349b4f`. The table states every generating axis, its unfavorable coordinate and its forced coordinate. Their full row-level Cartesian product is preserved in `claims/SFT-INFO-NOISE-ERROR-001/candidate_census.json`.

Structural axis	Rejected alternative	Exact rejection	Forced coordinate
`source`	`partial-source-support`	An omitted source hides a possible predecessor.	`complete-canonical-sources` - Every registered clean word occurs once.
`received`	`sampled-received-forms`	Sampling cannot establish the transformation boundary complete.	`complete-registered-images` - Every generated image under registered actions is retained.
`action`	`opaque-perturbation`	An opaque change hides which source produced which image.	`held-action-relation` - Each cell retains source, image and action identity.
`error`	`scalar-error-size`	A scalar count erases changed positions and labels.	`complete-changed-position-trace` - Every differing position retains old and new labels; identity is empty One.
`detection`	`probability-threshold`	A threshold imports a distribution and free parameter.	`invalid-codeword-witness` - A changed received form outside valid code support is detected structurally.

Structural axis	Rejected alternative	Exact rejection	Forced coordinate
`predecessors`	`chosen-likely-source`	Choosing one predecessor imports a prior.	`complete-source-class` - Every registered source mapping to the received form is retained.
`correction`	`nearest-form-assumption`	Nearest requires an unforced metric and can tie.	`singleton-predecessor` - Correction occurs exactly when one registered source remains.
`cause`	`stochastic-noise-source`	A random cause is neither generated nor needed.	`deterministic-transformation-family` - Noise names the complete registered action relation over deterministic forms.
`generality`	`sampled-masks`	Examples do not establish a fresh action image.	`action-successor` - A fresh registered action adds its exact source/image cells and updates predecessor classes.
`addition`	`extra-error-rate-or-metric`	A rate, distribution or metric is not supplied by exact transformations.	`no-extra-error-model` - Detection and correction use only code support and predecessor identity.

12.3 Exhaustion, unique survivor and laws

> *The noise/error kernel is complete source and received support, held action provenance, exact position traces, invalid-code detection, complete predecessor classes, singleton correction and no stochastic or metric rule.*

The engine recorded 1,024 decisions, 1,023 eliminations and exactly one survivor, `complete-canonical-sources__complete-registered-images__held-action-relation__complete-changed-position-trace__invalid-codeword-witness__complete-source-class__singleton-predecessor__deterministic-transformation-family__action-successor__no-extra-error-model`. Replacing any survivor coordinate with the rejected coordinate triggers the table's failure condition; the extra-rule coordinate prevents an unforced addition. This proves minimality and named-shape uniqueness inside the frozen grammar. The survivor entails:

- identity transport has structural empty-One error trace.
- nonidentity error retains every changed position and both labels.
- detection requires a received form outside the valid codebook.
- exact correction is equivalent to a singleton registered predecessor class.
- ambiguous received forms preserve all source predecessors rather than choosing a likely one.

12.4 Operational demonstrations and adverse controls

- `identity-error` - Identity transport returns structural empty-One error. Result: PASS.
- `position-trace` - The first-label change retains its exact position and labels. Result: PASS.
- `detection` - A one-label image outside the repetition codebook is structurally detectable. Result: PASS.
- `unique-correction` - The registered b/a/a image has exactly source a. Result: PASS.
- `ambiguous-control` - The shared b/a/b image retains both predecessors and refuses unique correction. Result: PASS.

The engine-level unfavorable controls are:

- `false_premise` - expected: reject a generated form missing the first required structural condition Observed: An omitted source hides a possible predecessor. Result: PASS; receipt `sha256:aa0a31d6b2f952723af21b700fd42b6e3524208149527bf5e2a28a1ef1db1f4f`.
- `tampered_source` - expected: reject a changed official source identity Observed: the changed identity differs from the registered source manifest Result: PASS; receipt `sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26`.
- `tampered_artifact` - expected: reject a missing, duplicated or additional survivor Observed: the generated product contains exactly one registered survivor Result: PASS; receipt `sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219`.
- `boundary` - expected: halt any attempt to import an excluded object or answer-producing model Observed: no stochastic noise cause; no floating error rate or threshold; no assumed metric or nearest-neighbor rule; no unrecorded source

inference Result: PASS; receipt
`sha256:e9746338cf385cffd4197fa5fbf34f715cc9519738827c592e17a26421e0859f`.

12.5 Depth-independent closure

Base: One source under its identity action yields itself, empty-One error and a singleton predecessor class.

Successor: Adding one registered action adds one exact image cell for each source on which it acts. Error traces compare those forms positionwise, and each affected received predecessor class extends by precisely the new source cell.

The executed product closes the registered representation alternatives. The base/successor certificate extends that rule to every generated finite support size, word depth, source extension or action extension named by the claim. It does not convert finite generation into a completed infinity.

12.6 Meaning and scientific consequence

Noise is not assumed to be random. It is the complete registered family of source-to-received transformations, each retaining the source, received form and action identity. An error is the exact changed-position trace, including the previous and received labels. Detection follows when a nonidentity image leaves valid code support. Correction follows only when the received form has one registered source predecessor. If several sources remain, the machine returns their whole class and refuses a likely, nearest or prior-weighted choice.

12.7 Correspondence only after sealing

After the SFT survivor and receipt are fixed, the following established terms may identify translation points: noise, error pattern, error detection, syndrome, decoding ambiguity. They are not dependencies and did not provide formulas, thresholds or answer tables. A mismatch remains visible as a boundary result rather than being repaired by changing the sealed law.

12.8 Exact exclusions and limitation

- no stochastic noise cause.
- no floating error rate or threshold.
- no assumed metric or nearest-neighbor rule.
- no unrecorded source inference.

This theorem closes finite structural noise and error. It does not assign a stochastic error frequency, physical noise mechanism or asymptotic reliability rate.

12.9 Machine evidence identities

- Registered status: `independently_replicated`.
- Source manifest: `sha256:84c7f7977b3822e236f0b3710e5a7a01465c4024aaa58b6a847bca4c1adcc410`.
- Derivation seal: `sha256:d3852e47b6d6b9aaa1328367c912800ffca2d78396fb432d0e56b3c5dc66c744`.
- Independent implementation: `sha256:b07e2fda2a8b23a887a84ace1c0cf08613a2802ccbb98fc7f20500ed962532b5`.
- Independent certificate: `sha256:f5c174a125958108183e7b9ea19ad76cdf799784b6b4ca9a32aa9bd44deff50d`.
- External validation: `sha256:8dba26a7c7b76d11f393d114625f8ffd0486e1b8c3c9bda9f71c44d096cbb92e`.
- Engine receipt: `sha256:bc42a3ab0291a59b278654bed369bb1d64b6abe1f9278af9f89eac78211f48b9`.
- Engine receipt path: `receipts/engine/model_admitted/SFT-INFO-NOISE-ERROR-001-bc42a3ab0291a59b.json`.

13. Derivation 8: Exact finite coding, detection and correction

Claim identity: `SFT-INFO-CODING-001`

13.1 Question, necessity and theorem

Coding theory must establish correction from complete structural error alternatives. A metric or probability may summarize a sealed code later but cannot select which received forms decode uniquely.

The exact theorem statement is:

> *A finite code is a nonempty held selection of complete generated equal-position word support plus a total source/codeword relation. Codeword difference is the exact changed-position trace. A registered error family is detectable when every nonidentity image lies outside the codebook and correctable when distinct codewords have disjoint complete error-image supports. Decoding is then the unique source whose image family contains the received word. Redundancy is admitted only when those retained positions force separation.*

Admitted dependencies: `SFT-FOUNDATION-FOLD-ASSEMBLY-001`, `SFT-MATH-COMBINATORICS-001`, `SFT-INFO-COMPRESSION-REDUNDANCY-001`, `SFT-INFO-NOISE-ERROR-001`.

13.2 Derivation chain, grammar and boundary

Fold assembly supplies word support; noise supplies registered transformations and predecessor classes; compression supplies lawful redundancy; combinatorics exhausts image families. Eleven axes force finite detection and correction through exact support separation.

Generation rule: Generate the complete product of word support, codebook selection, source encoding, position identity, difference trace, error-family coverage, detection, correction, redundancy witness, generality and extra-code status.

Exact grammar boundary: All finite codes generated from complete held-label word support, held codebooks, exact source maps, common position carriers, registered error actions and complete error-image families.

The complete product contains 2,048 candidates. Its completeness-certificate identity is `sha256:b1ec186cbef105762fbf52baed16e644e0f2f0944e6aa739f3b4da39f160a368`. The table states every generating axis, its unfavorable coordinate and its forced coordinate. Their full row-level Cartesian product is preserved in `claims/SFT-INFO-CODING-001/candidate_census.json`.

Structural axis	Rejected alternative	Exact rejection	Forced coordinate
`support`	`partial-word-support`	Omission hides possible code or error images.	`complete-generated-word-support` - Every held-label word over the registered positions occurs once.
`codebook`	`free-valid-list`	A free list lacks a complete source support boundary.	`held-support-selection` - The codebook is an exact nonempty selection of generated words.
`encoding`	`implicit-assignment`	An implicit assignment erases source provenance.	`total-source-code-relation` - Every source message has one retained codeword.
`positions`	`unequal-or-unlabelled-positions`	Position changes cannot be located on unequal or anonymous carriers.	`common-canonical-position-trace` - Every word uses the same retained ordered position carrier.
`difference`	`scalar-distance`	A scalar erases which positions and labels differ.	`complete-difference-trace` - Every changed position retains both code labels; equality is empty One.
`errors`	`sampled-error-outputs`	Sampling cannot prove image supports disjoint.	`complete-registered-error-images` - Every registered action image of every codeword is generated.
`detection`	`likely-corruption`	Likelihood imports a prior and threshold.	`noncodeword-image` - Every nonidentity registered image lies outside the valid codebook.
`correction`	`nearest-codeword`	A metric can tie and is not structurally supplied.	`disjoint-error-image-support` - Every received image belongs to at most one source codeword family.
`redundancy`	`unaccounted-padding`	Padding without a correction witness is an unforced addition.	`separation-forcing-positions` - Every added position participates in the exact disjoint-image certificate.

Structural axis	Rejected alternative	Exact rejection	Forced coordinate
`generality`	`fixed-mask-example`	One mask does not establish all registered actions.	`error-action-successor` - A fresh action extends every codeword image family and disjointness is rechecked.
`addition`	`extra-distance-or-rate-rule`	A borrowed metric or asymptotic rate can select the code.	`no-extra-coding-rule` - Code status follows only from generated support and error images.

13.3 Exhaustion, unique survivor and laws

> *The coding kernel is complete word support, held codebook and source map, common positions, exact differences, complete registered error images, structural detection, disjoint-support correction and witnessed redundancy.*

The engine recorded 2,048 decisions, 2,047 eliminations and exactly one survivor, `complete-generated-word-support__held-support-selection__total-source-code-relation__common-canonical-position-trace__complete-difference-trace__complete-registered-error-images__noncodeword-image__disjoint-error-image-support__separation-forcing-positions__error-action-successor__no-extra-coding-rule`. Replacing any survivor coordinate with the rejected coordinate triggers the table's failure condition; the extra-rule coordinate prevents an unforced addition. This proves minimality and named-shape uniqueness inside the frozen grammar. The survivor entails:

- valid codewords are a held selection of complete generated word support.
- difference retains every changed canonical position and both labels.
- detection requires all nonidentity registered images outside the codebook.
- correction is equivalent to pairwise disjoint complete error-image supports.
- a correctable received form decodes by exact unique source membership without a prior.

13.4 Operational demonstrations and adverse controls

- `support` - Two labels over three positions generate eight unique words. Result: PASS.
- `difference` - The repetition codewords differ at all three retained positions. Result: PASS.
- `detection` - Every single-label change of a width-three repetition codeword lies outside the codebook. Result: PASS.
- `correction` - The two complete single-label error-image supports are disjoint. Result: PASS.
- `decoding` - A received b/a/a word belongs uniquely to the all-a source family. Result: PASS.
- `unprotected-control` - The one-position uncoded family cannot correct a single label change. Result: PASS.

The engine-level unfavorable controls are:

- `false_premise` - expected: reject a generated form missing the first required structural condition Observed: Omission hides possible code or error images. Result: PASS; receipt `sha256:f7172e9412e2f058233eedc9292241ada463109f88a567019995e1f842fa7acb`.
- `tampered_source` - expected: reject a changed official source identity Observed: the changed identity differs from the registered source manifest Result: PASS; receipt `sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26`.
- `tampered_artifact` - expected: reject a missing, duplicated or additional survivor Observed: the generated product contains exactly one registered survivor Result: PASS; receipt `sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219`.
- `boundary` - expected: halt any attempt to import an excluded object or answer-producing model Observed: no stochastic error model; no floating distance or likelihood threshold; no unaccounted redundancy; no asymptotic rate imported as a finite proof Result: PASS; receipt `sha256:dbc8aa5bf22488e77f3b48616b7f6856c06cc2ed7350c94e878078caaed92030`.

13.5 Depth-independent closure

Base: One registered error action generates one image per codeword; detection and support disjointness are exhaustively checked.

Successor: Adding one error action appends its exact image to every affected codeword family. Detection checks valid-word membership and correction checks every newly possible intersection with all other source families.

The executed product closes the registered representation alternatives. The base/successor certificate extends that rule to every generated finite support size, word depth, source extension or action extension named by the claim. It does not convert finite generation into a completed infinity.

13.6 Meaning and scientific consequence

Coding theory follows from complete error-image separation. A codebook is a held selection of generated equal-position words, and every registered error action is applied to every codeword. Detection requires nonidentity images outside the codebook; correction requires disjoint complete image supports between distinct codewords. Width-three repetition is demonstrated by exhausting all single-label images, while an unprotected one-position family supplies the adverse control. No distance metric or error probability selects the code; a conventional distance may summarize the sealed difference traces afterward.

13.7 Correspondence only after sealing

After the SFT survivor and receipt are fixed, the following established terms may identify translation points: block code, Hamming distance, error detection, error correction, repetition code. They are not dependencies and did not provide formulas, thresholds or answer tables. A mismatch remains visible as a boundary result rather than being repaired by changing the sealed law.

13.8 Exact exclusions and limitation

- no stochastic error model.
- no floating distance or likelihood threshold.
- no unaccounted redundancy.
- no asymptotic rate imported as a finite proof.

This theorem closes exact finite classical coding for any explicitly registered error family. Quantum codes, fault tolerance and unbounded error thresholds belong to later branches.

13.9 Machine evidence identities

- Registered status: `independently_replicated`.
- Source manifest: `sha256:6e8db3894be5bea7163c2a18dbffa6879fa56308a2b3d3fb49a56f0d98c920e5`.
- Derivation seal: `sha256:357b7cd32f117b55c0bad9606247bf12db72c82b999710d1c38698a5d895523d`.
- Independent implementation: `sha256:a7b52a43add0bb3b0e84de51a71a55320eba8beae57793ef328833b89f6427c1`.
- Independent certificate: `sha256:8502145761cfda22f034d5ca13deca41128060420454b3d93f689bd2d822e7f7`.
- External validation: `sha256:d1be5779eb0de948cc041a05e504a3dc9bb66fba0985f16bf5ed8d015fdc6a66`.
- Engine receipt: `sha256:7c7648bdba686025c8b1f647b3e63a242da24234aa953cac184021f878b07711`.
- Engine receipt path: `receipts/engine/model_admitted/SFT-INFO-CODING-001-7c7648bdba686025.json`.

14. Derivation 9: Exact mutual and conditional information from joint observation incidence

Claim identity: `SFT-INFO-MUTUAL-CONDITIONAL-001`

14.1 Question, necessity and theorem

Conditional and mutual information must preserve the observation cells that create the relation. Collapsing them into an imported logarithmic scalar would discard the exact structural proof.

The exact theorem statement is:

> *Conditional information is the complete exact restriction of one total observation to every class of another. Mutual dependence is the complete joint-incidence ledger comparing each observed joint support part with the product of its exact marginal parts; absent cells remain structural empty One and no signed subtraction is used. Factorization holds exactly when every joint cell is present and its part equals the marginal product. Thus conditioning and dependence are derived from deterministic support and observation, without priors, logarithms, stochastic causes or floating values.*

Admitted dependencies: `SFT-MATH-PROBABILITY-STATISTICS-001`, `SFT-MATH-COMBINATORICS-001`, `SFT-INFO-ENTROPY-UNCERTAINTY-001`, `SFT-INFO-QUANTITY-001`.

14.2 Derivation chain, grammar and boundary

Probability supplies exact support parts, entropy supplies observation classes and quantity supplies retained distinctions. Ten axes force conditional restriction, joint incidence and exact factorization.

Generation rule: Generate the complete product of support, two observations, marginal classes, joint incidence, conditional restriction, dependence comparison, factorization, deterministic status, generality and extra-measure status.

Exact grammar boundary: All finite mutual and conditional information structures generated from complete deterministic support, two total observation maps, their exact marginal classes, complete joint cells and exact positive part relations.

The complete product contains 1,024 candidates. Its completeness-certificate identity is `sha256:b11ba0d7ab94f6fa10d3651be8f6c6935d4c4a70787d3ff51648695c1c477ea3`. The table states every generating axis, its unfavorable coordinate and its forced coordinate. Their full row-level Cartesian product is preserved in `claims/SFT-INFO-MUTUAL-CONDITIONAL-001/candidate_census.json`.

Structural axis	Rejected alternative	Exact rejection	Forced coordinate
`support`	`partial-source-support`	Omission changes marginal, joint and conditional parts.	`complete-deterministic-support` - Every registered source form occurs exactly once.
`observations`	`partial-or-multivalued-views`	An incomplete or multivalued view cannot form exact classes.	`two-total-observation-maps` - Each source form has exactly one image in each view.
`marginals`	`named-marginal-scores`	Scores erase which source forms produce each image.	`exact-provenance-classes` - Each image retains its complete source class and exact whole part.
`joint`	`sampled-joint-cells`	Sampling can hide absent or repeated combinations.	`complete-joint-incidence-ledger` - Every pair of marginal labels has an explicit member class or empty One.
`conditional`	`renormalized-prior`	A prior or guessed normalization is not generated.	`exact-class-restriction` - Each target class is intersected with the complete held given class.
`mutual`	`scalar-log-ratio`	A log ratio imports real values and erases the cells responsible.	`joint-versus-product-ledger` - Every joint part is compared exactly with its marginal product while retaining provenance.
`factorization`	`uncorrelated-assertion`	A label does not establish every joint cell and weight.	`all-cell-exact-equality` - Every joint cell exists and equals its exact marginal product.
`cause`	`stochastic-source-premise`	A random cause is neither forced nor required.	`deterministic-observation-incidence` - Dependence is a relation among exact observation classes over deterministic support.
`generality`	`small-table-only`	A table does not establish the next source update.	`source-successor-update` - A fresh source enters one marginal class and one joint cell, updating exact parts.
`addition`	`extra-base-prior-or-scalar`	The added choice is a free parameter.	`no-extra-information-measure` - The ledgers follow only from complete support and total observations.

14.3 Exhaustion, unique survivor and laws

> *The mutual/conditional kernel is complete deterministic support, two total observations, exact marginal and joint classes, exact conditional restrictions, provenance-retaining joint/product comparison, all-cell factorization, successor closure and no stochastic or logarithmic measure.*

The engine recorded 1,024 decisions, 1,023 eliminations and exactly one survivor, `complete-deterministic-support__two-total-observation-maps__exact-provenance-classes__complete-joint-incidence-ledger__exact-class-restriction__joint-versus-product-ledger__all-cell-exact-equality__deterministic-observation-incidence__source-successor-update__no-extra-information-measure`. Replacing any survivor coordinate with the rejected coordinate triggers the table's failure condition; the extra-rule coordinate prevents an unforced addition. This proves minimality and named-shape uniqueness inside the frozen grammar. The survivor entails:

- conditional rows partition every nonempty given class by target observation.
- every conditional part is exact and positive; absent intersections are empty One.
- the joint ledger explicitly contains every marginal-label pair cell.
- factorization holds exactly when every joint cell equals its marginal-part product.
- dependence retains the responsible joint cells and comparison direction without signed subtraction.

14.4 Operational demonstrations and adverse controls

- `conditional-parts` - Conditioning the second coordinate on the first yields two exact half-parts in each given class. Result: PASS.
- `factorized-grid` - The complete two-by-two coordinate grid factorizes exactly. Result: PASS.
- `empty-dependence` - A factorized grid has structural empty-One dependence ledger. Result: PASS.
- `resolved-copy` - An observation is conditionally resolved by an exact copy of itself. Result: PASS.
- `dependent-copy` - Two identical nontrivial views do not factorize and retain their responsible cells. Result: PASS.

The engine-level unfavorable controls are:

- `false_premise` - expected: reject a generated form missing the first required structural condition Observed: Omission changes marginal, joint and conditional parts. Result: PASS; receipt `sha256:f564b6aa6b8ba61accc8d69530492f79dc2980f78cc08304bea6d57f9801d75d`.
- `tampered_source` - expected: reject a changed official source identity Observed: the changed identity differs from the registered source manifest Result: PASS; receipt `sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26`.
- `tampered_artifact` - expected: reject a missing, duplicated or additional survivor Observed: the generated product contains exactly one registered survivor Result: PASS; receipt `sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219`.
- `boundary` - expected: halt any attempt to import an excluded object or answer-producing model Observed: no logarithmic mutual-information proof value; no signed or floating dependence score; no stochastic cause or prior distribution; no incomplete joint census presented as factorization Result: PASS; receipt `sha256:23dc6a258a994a86504934396406af14c8c3d49dbbe6eaf38d4864f3f45aca3b`.

14.5 Depth-independent closure

Base: One source form gives one whole marginal cell, one whole joint cell and one whole conditional cell.

Successor: Adding one fresh source form places it in exactly one class of each observation and therefore one joint cell; all marginal, joint and conditional counts and exact parts are recomputed from retained membership.

The executed product closes the registered representation alternatives. The base/successor certificate extends that rule to every generated finite support size, word depth, source extension or action extension named by the claim. It does not convert finite generation into a completed infinity.

14.6 Meaning and scientific consequence

Conditional information is exact restriction: within every class of a given observation, the target observation partitions the retained microstates and supplies exact positive parts. Mutual dependence is kept as a complete joint-incidence ledger. Every joint cell records members, exact joint part, exact marginal parts, their product and the comparison relation; an absent cell is empty One. Factorization holds only when every cell is present and exactly matches the marginal product. This avoids signed subtraction and logarithmic ratios while retaining all evidence needed to identify where dependence occurs.

14.7 Correspondence only after sealing

After the SFT survivor and receipt are fixed, the following established terms may identify translation points: conditional information, mutual information, joint distribution, marginal, independence. They are not dependencies and did not provide formulas, thresholds or answer tables. A mismatch remains visible as a boundary result rather than being repaired by changing the sealed law.

14.8 Exact exclusions and limitation

- no logarithmic mutual-information proof value.
- no signed or floating dependence score.
- no stochastic cause or prior distribution.
- no incomplete joint census presented as factorization.

This closes exact finite structural conditional information and dependence. Conventional scalar Shannon mutual information is a post-seal correspondence summary, not an SFT proof value.

14.9 Machine evidence identities

- Registered status: `independently_replicated`.
- Source manifest: `sha256:f8ea0d5e7e10a01d084c85bd9cff114e02014a63432f38483131101c8c0b2982`.
- Derivation seal: `sha256:34061aeffca9322a79eedb4a54ca2755ab8959dd0894e432e747f56be3f443f1`.
- Independent implementation: `sha256:9ab64f13d6feff5206edd252cece25cc802249f6d39064d3bfd9c39defa7138d`.
- Independent certificate: `sha256:ebc36a627bb287d5108dd3714f18855e44ffd50b871fdb82e91235506ae8dfd9`.
- External validation: `sha256:390455380b90b13b9cca82426eefca22b65315665c69dceeaa64c056e2089217`.
- Engine receipt: `sha256:c311f27de12d70ea2307acb2d5b23f83cb065b6b28fe5c08ac4742f6f1aadc26`.
- Engine receipt path: `receipts/engine/model_admitted/SFT-INFO-MUTUAL-CONDITIONAL-001-c311f27de12d70ea.json`.

15. Derivation 10: Exact information conservation, loss and transformation

Claim identity: `SFT-INFO-CONSERVATION-LOSS-001`

15.1 Question, necessity and theorem

Information conservation and loss require pair-level accounting: image counts alone cannot show which source alternatives remain recoverable or what record reversal requires.

The exact theorem statement is:

> *An information transformation is a total single-valued map over complete source support. Every source distinction is conserved as an image distinction or closed inside an exact predecessor fibre; these two ledgers are disjoint and exhaustive. Reversibility is forced exactly by singleton fibres. An irreversible transformation can be reversed only when a retained predecessor label resolves each non-singleton fibre. Under composition, downstream retained distinctions are a subset of those retained upstream, so a map creates no unrecorded source distinction.*

Admitted dependencies: `SFT-MATH-DYNAMICAL-SYSTEMS-001`, `SFT-MATH-LOGIC-PROOF-001`, `SFT-INFO-ENCODING-DECODING-001`, `SFT-INFO-CHANNEL-CAPACITY-001`, `SFT-INFO-MUTUAL-CONDITIONAL-001`.

15.2 Derivation chain, grammar and boundary

Encoding supplies inverse classes, channels supply relations, mutual information supplies joint incidence and dynamics supplies composition. Ten axes force exhaustive pair accounting and the compositional loss law.

Generation rule: Generate the complete product of source support, totality, provenance, pair accounting, fibres, loss, reversibility, composition, generality and extra-source status.

Exact grammar boundary: All finite deterministic information transformations generated from complete canonical source support, total single-valued maps, exact image fibres, complete source-pair ledgers and transformation composition.

The complete product contains 1,024 candidates. Its completeness-certificate identity is `sha256:ad0454f13ea0f144524fa1e94dc4d23a7cae305a1d392ba12f2bfe7dbff59923`. The table states every generating axis, its unfavorable coordinate and its forced coordinate. Their full row-level Cartesian product is preserved in `claims/SFT-INFO-CONSERVATION-LOSS-001/candidate_census.json`.

Structural axis	Rejected alternative	Exact rejection	Forced coordinate
`support`	`partial-input-support`	Omission hides distinctions that may be lost.	`complete-canonical-source` - Every registered source form occurs once.
`map`	`partial-or-multivalued-rule`	An incomplete or multivalued rule cannot account for every source.	`total-single-valued-map` - Each source has exactly one held image.
`provenance`	`image-only-output`	Image-only output erases which sources entered each result.	`exact-predecessor-fibres` - Every image retains its complete source class.
`accounting`	`output-cardinality-only`	A count erases which distinctions survive.	`complete-source-pair-partition` - Every source pair is classified exactly as retained or closed.
`loss`	`lower-score-assertion`	A score does not identify unavailable predecessors.	`non-singleton-image-fibre` - Loss is the exact source distinctions closed inside an image fibre.
`reversal`	`assumed-inverse`	An inverse cannot choose among merged predecessors.	`singleton-fibre-or-held-record` - Singleton fibres reverse directly; merged fibres require retained predecessor labels.
`composition`	`fresh-downstream-distinction`	A deterministic image map has no source for a distinction already closed.	`retained-pair-subset` - A downstream map can preserve or close an upstream distinction but cannot reopen it.
`conservation`	`unchanged-scalar`	An unchanged scalar can hide exchanges among distinctions.	`retained-plus-closed-exhaustion` - The two exact ledgers disjointly exhaust all source distinctions.
`generality`	`sampled-transformations`	Examples do not prove a fresh source image accounted.	`source-successor-map-cell` - A fresh source adds one held map cell and all new pair comparisons.
`addition`	`unrecorded-created-distinction`	A new distinction without source or record violates provenance.	`no-extra-information-source` - All output distinctions trace to separated source images or declared external input.

15.3 Exhaustion, unique survivor and laws

> *The transformation kernel is complete source support, a total held map, exact predecessor fibres, exhaustive retained/closed source distinctions, singleton-fibre reversibility, held reverse records, compositional retention monotonicity and no unrecorded information source.*

The engine recorded 1,024 decisions, 1,023 eliminations and exactly one survivor, `complete-canonical-source__total-single-valued-map__exact-predecessor-fibres__complete-source-pair-partition__` `non-singleton-image-fibre__singleton-fibre-or-held-record__retained-pair-subset__retained-plus-closed-exhaustion__source-successor-map-cell__no-extra-information-source`. Replacing any survivor coordinate with the rejected coordinate triggers the table's failure condition; the extra-rule coordinate prevents an unforced addition. This proves minimality and named-shape uniqueness inside the frozen grammar. The survivor entails:

- retained and closed ledgers are disjoint and exhaust every source distinction.
- information loss is exactly the source distinctions inside non-singleton image fibres.
- a transformation is reversible exactly when every predecessor fibre is singleton.
- a non-singleton fibre requires a retained predecessor label for exact reconstruction.
- composition cannot restore a distinction already closed by an upstream deterministic map.

15.4 Operational demonstrations and adverse controls

- `identity-conservation` - Identity retains all six source distinctions and closes none. Result: PASS.
- `prefix-loss` - Prefix projection retains four cross-prefix distinctions and closes two exact predecessor pairs. Result: PASS.
- `reversibility-boundary` - Identity is reversible while prefix projection is not. Result: PASS.
- `reverse-record` - Each prefix image retains its two possible source labels. Result: PASS.
- `data-processing` - Erasing the retained prefix cannot reopen either previously closed distinction. Result: PASS.

The engine-level unfavorable controls are:

- `false_premise` - expected: reject a generated form missing the first required structural condition Observed: Omission hides distinctions that may be lost. Result: PASS; receipt `sha256:e49483a81fecb2b86c78c6d275a16553a1d404e42699dcfd48c68e3cc368ce89`.
- `tampered_source` - expected: reject a changed official source identity Observed: the changed identity differs from the registered source manifest Result: PASS; receipt `sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26`.
- `tampered_artifact` - expected: reject a missing, duplicated or additional survivor Observed: the generated product contains exactly one registered survivor Result: PASS; receipt `sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219`.
- `boundary` - expected: halt any attempt to import an excluded object or answer-producing model Observed: no image without source provenance; no assumed inverse across merged predecessors; no scalar conservation law replacing distinction ledgers; no downstream reopening without a declared external record Result: PASS; receipt `sha256:985dfeb5e8a4ab2440265fb290f9431a0a8e901052f776a68ff919c0b2fd65f6`.

15.5 Depth-independent closure

Base: One source maps to one held image, has one singleton fibre and structural empty-One pair ledgers.

Successor: Adding one fresh source adds one image membership and one pair against each prior source. Image equality places each new pair exactly in the retained or closed ledger and updates one predecessor fibre.

The executed product closes the registered representation alternatives. The base/successor certificate extends that rule to every generated finite support size, word depth, source extension or action extension named by the claim. It does not convert finite generation into a completed infinity.

15.6 Meaning and scientific consequence

Information conservation is pair accounting. A total map classifies every source distinction as retained by unequal images or closed inside one predecessor fibre. Those ledgers are disjoint and exhaustive. Reversibility is equivalent to singleton fibres; a merged fibre requires a retained predecessor label. Under composition, the downstream retained pair set is a subset of the upstream set, so a deterministic map cannot reopen an unrecorded distinction. Conservation therefore does not mean an opaque scalar remained constant: it means every source alternative has an exact disposition.

15.7 Correspondence only after sealing

After the SFT survivor and receipt are fixed, the following established terms may identify translation points: data processing, information loss, reversible map, many-to-one map, sufficient record. They are not dependencies and did not provide formulas, thresholds or answer tables. A mismatch remains visible as a boundary result rather than being repaired by changing the sealed law.

15.8 Exact exclusions and limitation

- no image without source provenance.
- no assumed inverse across merged predecessors.
- no scalar conservation law replacing distinction ledgers.
- no downstream reopening without a declared external record.

This theorem closes exact finite deterministic information transformation. Thermodynamic energy costs and operational quantum measurement belong to later physical and quantum branches.

15.9 Machine evidence identities

- Registered status: `independently_replicated`.
- Source manifest: `sha256:60ac768f9abc68df9e028bcd206af56b72e1de369257164082dbf31d0bb92c61`.
- Derivation seal: `sha256:9c9e86f6923115194a041aea4ee7d8c49133e1653f5f60a17e809c782dbe2517`.
- Independent implementation: `sha256:d13db48f1136b543655727389a65b1c1e0dacf8cdcfaae856163ef3ffb36d051`.
- Independent certificate: `sha256:5b195602b942201b80b630c2745f56f5bb1998039039e358240583c72005c8c9`.
- External validation: `sha256:11ccba3a43e1918ab7a42d9ce2437b65c11c8f37e12b82bfd420be0655c1755e`.
- Engine receipt: `sha256:4e2892e85c1624a2234163fdd51377a7086219ce0945cbf6acfcaa41664303f4`.
- Engine receipt path: `receipts/engine/model_admitted/SFT-INFO-CONSERVATION-LOSS-001-4e2892e85c1624a2.json`.

16. Derivation 11: Exact classical and probabilistic information correspondence

Claim identity: `SFT-INFO-CLASSICAL-PROBABILISTIC-001`

16.1 Question, necessity and theorem

The relation between classical and probabilistic information must be explicit in a deterministic theory. Observation classes derive exact probability while preserving every underlying state.

The exact theorem statement is:

> *Classical information is one held member of a complete deterministic support with its provenance retained. Probabilistic information is the exact observation-class ledger over that same support: every class retains its deterministic microstates and exact positive part of the whole. Observation resolves a classical state only for a singleton class; otherwise it returns the complete unresolved class. Deterministic transformations push the ledger forward by exact predecessor grouping. Probability therefore describes unavailable distinctions and does not introduce stochastic dynamics, floating propensities, numerical zero or priors.*

Admitted dependencies: `SFT-MATH-PROBABILITY-STATISTICS-001`, `SFT-INFO-ENTROPY-UNCERTAINTY-001`, `SFT-INFO-MUTUAL-CONDITIONAL-001`,

`SFT-INFO-CONSERVATION-LOSS-001`.

16.2 Derivation chain, grammar and boundary

Mathematical probability supplies exact parts, entropy supplies observation classes, mutual information supplies joint restriction and conservation supplies deterministic pushforward. Ten axes force one shared support account.

Generation rule: Generate the complete product of deterministic support, classical state, observation, exact class parts, probabilistic meaning, resolution, transformation, whole closure, generality and extra-random status.

Exact grammar boundary: All finite classical/probabilistic information descriptions generated from complete deterministic support, one held classical state, total observation classes, exact positive support parts and total deterministic maps.

The complete product contains 1,024 candidates. Its completeness-certificate identity is `sha256:e03cac175631127fde7130ed62fb48d555e106539d14910987ceeece484ee25f`. The table states every generating axis, its unfavorable coordinate and its forced coordinate. Their full row-level Cartesian product is preserved in `claims/SFT-INFO-CLASSICAL-PROBABILISTIC-001/candidate_census.json`.

Structural axis	Rejected alternative	Exact rejection	Forced coordinate
`support`	`sampled-or-duplicated-support`	Sampling or duplication changes every exact part.	`complete-canonical-support` - Every deterministic microstate occurs once.
`classical`	`unheld-symbol-name`	A name alone does not locate a state in complete support.	`one-held-supported-state` - One exact support member is retained with its provenance.
`observation`	`free-weight-list`	Weights without microstates import a distribution.	`total-observation-classes` - Every microstate belongs to exactly one observed class.
`parts`	`floating-propensities`	Floating propensities import precision and causal assumptions.	`exact-class-whole-parts` - Each nonempty class count is retained relative to complete support count.
`meaning`	`ontic-random-cause`	A random cause is neither generated nor needed.	`unresolved-deterministic-distinction` - Each probabilistic class explicitly retains the deterministic alternatives observation closes.
`resolution`	`likely-state-selection`	Selecting the likely state imports a prior.	`singleton-class-only` - Only one-member classes resolve; larger classes remain complete.
`transformation`	`stochastic-transition-matrix`	A free transition matrix adds parameters and random dynamics.	`deterministic-predecessor-grouping` - A total map groups exact predecessor support into image classes.
`normalization`	`fitted-normalization`	A fitted factor can conceal omitted states.	`partition-exact-whole` - Disjoint complete class parts reconstruct the exact whole.
`generality`	`finite-examples-only`	Examples do not prove the next microstate update.	`microstate-successor` - A fresh microstate enters one class and every exact part is recomputed from membership.
`addition`	`extra-seed-prior-or-kernel`	The addition is a free parameter and may select results.	`no-extra-stochastic-law` - All probabilistic information follows from support, observation and deterministic maps.

16.3 Exhaustion, unique survivor and laws

> *The classical/probabilistic kernel is complete deterministic support, one held classical state, total observation classes, exact positive class parts, singleton-only resolution, deterministic pushforward, exact-whole closure, successor closure and no stochastic law.*

The engine recorded 1,024 decisions, 1,023 eliminations and exactly one survivor, `complete-canonical-support__one-held-supported-state__total-observation-classes__exact-class-whole-parts__unresolved-deterministic-distinction__singleton-class-only__deterministic-predecessor-grouping__partition-exact-whole__microstate-successor__no-extra-stochastic-law`. Replacing any survivor coordinate with the rejected coordinate triggers the table's failure condition; the extra-rule coordinate prevents an unforced addition. This proves minimality and named-shape uniqueness inside the frozen grammar. The survivor entails:

- a classical information state is one held member of complete support.

- a probabilistic row is an exact nonempty observation class and positive whole part.
- the complete class ledger reconstructs the exact whole.
- observation selects a state only when its retained class is singleton.
- deterministic pushforward groups predecessors without introducing random transitions.

16.4 Operational demonstrations and adverse controls

- `classical-held-state` - A held aa state retains its complete four-state provenance. Result: PASS.
- `exact-half-classes` - Prefix observation produces two exact one-over-two deterministic classes. Result: PASS.
- `exact-unequal-parts` - A three-member class and one-member class retain exact three-over-four and one-over-four parts. Result: PASS.
- `resolution-boundary` - Fine observation resolves aa while prefix observation retains the unresolved aa/ab class. Result: PASS.
- `pushforward-whole` - Deterministic prefix pushforward retains both predecessors and reconstructs the exact whole. Result: PASS.
- `deterministic-realization` - Every probabilistic prefix row exposes its deterministic member trace. Result: PASS.

The engine-level unfavorable controls are:

- `false_premise` - expected: reject a generated form missing the first required structural condition Observed: Sampling or duplication changes every exact part. Result: PASS; receipt `sha256:0132e22625c62b433b3964d5cbe1d88179a76521617716a231969f070854eee2`.
- `tampered_source` - expected: reject a changed official source identity Observed: the changed identity differs from the registered source manifest Result: PASS; receipt `sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26`.
- `tampered_artifact` - expected: reject a missing, duplicated or additional survivor Observed: the generated product contains exactly one registered survivor Result: PASS; receipt `sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219`.
- `boundary` - expected: halt any attempt to import an excluded object or answer-producing model Observed: no ontic stochastic transition premise; no floating or fitted probability; no numerical zero probability row; absence is empty One; no prior-based selection inside a non-singleton class Result: PASS; receipt `sha256:74818e0565fb579afa88110c99f2f9434067dfd932e1cf70649dd060b5e5128b`.

16.5 Depth-independent closure

Base: One deterministic microstate is one held classical state and one exact whole observation class.

Successor: Adding one fresh microstate extends complete support and exactly one observation class. The class ledgers and their exact whole parts update from retained membership, while each individual state remains deterministic.

The executed product closes the registered representation alternatives. The base/successor certificate extends that rule to every generated finite support size, word depth, source extension or action extension named by the claim. It does not convert finite generation into a completed infinity.

16.6 Meaning and scientific consequence

Classical and probabilistic information are two resolutions of one deterministic carrier. A classical state is one held support member with its provenance. A probabilistic row is an exact nonempty observation class and its positive part of complete support. Observation identifies a state only when that class is singleton; otherwise all predecessors remain. A deterministic map pushes the ledger forward by grouping exact predecessor support. Thus probability records unavailable distinctions without adding a stochastic transition law, prior, seed or floating propensity.

16.7 Correspondence only after sealing

After the SFT survivor and receipt are fixed, the following established terms may identify translation points: classical state, ensemble, probability distribution, pushforward, epistemic uncertainty. They are not dependencies and did not provide formulas, thresholds or answer tables. A mismatch remains visible as a boundary result rather than being repaired by changing the sealed law.

16.8 Exact exclusions and limitation

- no ontic stochastic transition premise.
- no floating or fitted probability.
- no numerical zero probability row; absence is empty One.
- no prior-based selection inside a non-singleton class.

This theorem closes finite classical/probabilistic information correspondence. Empirical frequencies require separately registered blind data protocols and do not alter the formal law.

16.9 Machine evidence identities

- Registered status: `independently_replicated`.
- Source manifest: `sha256:30456ab6222e0f2cf74333305772e57515b2c1590739d8072d6393fed8364e5f`.
- Derivation seal: `sha256:1c937455ec2a12940b3d1521f5a88e3c21a1b51bb5f4b6d432f6498765a499f0`.
- Independent implementation: `sha256:e2c15ace1a24f33267a6d3d92e02b4cacd33c59853328561eff2869b0885c606`.
- Independent certificate: `sha256:185f98f5264a57295ac54685e839caf0f3bb9a4de766f1635d308d8563885763`.
- External validation: `sha256:11d9cf6b6aff03928993b4c343b0a01ec8a857acaf3f22c573a7fcda53d24ce3`.
- Engine receipt: `sha256:ce0515fdf69d368bddb313faad3a88043550e90e3df21c4049267e1a5e54fe7c`.
- Engine receipt path: `receipts/engine/model_admitted/SFT-INFO-CLASSICAL-PROBABILISTIC-001-ce0515fdf69d368b.json`.

17. Derivation 12: Exact quantum-information support correspondence

Claim identity: `SFT-INFO-QUANTUM-CORRESPONDENCE-001`

17.1 Question, necessity and theorem

Quantum information must first establish its exact alternative carrier and its relation to classical and probabilistic information. Operational quantum effects cannot be used to select that foundation.

The exact theorem statement is:

> *One Fold distinction supplies the native information unit. Across k canonical positions, the complete generated held-label word family supplies the exact branch support corresponding to a finite quantum basis support, and complete pair cells supply joint-state support. Calling this support superposition-equivalent asserts equality of enumerated alternatives and provenance only: amplitudes, phase and dynamics are not imported. Observation returns an exact retained branch class and a complete reconstruction record. Thus classical held states, probabilistic observation classes and quantum basis support share one exact information carrier at this boundary.*

Admitted dependencies: `SFT-FOUNDATION-FOLD-ASSEMBLY-001`, `SFT-MATH-CATEGORY-TYPE-COMPOSITION-001`, `SFT-INFO-QUANTITY-001`, `SFT-INFO-CONSERVATION-LOSS-001`, `SFT-INFO-CLASSICAL-PROBABILISTIC-001`.

17.2 Derivation chain, grammar and boundary

Fold assembly supplies words, information quantity supplies position units, conservation supplies records, classical/probabilistic correspondence supplies held states and observation classes, and composition supplies pair cells.

Generation rule: Generate the complete product of native unit, word support, basis correspondence, branch provenance, superposition boundary, joint support, observation record, reconstruction, generality and extra-quantum status.

Exact grammar boundary: All finite quantum-information support correspondences generated from the two held Fold labels, canonical word positions, complete branch support, bijective basis names, pair-cell composition and total observation classes.

The complete product contains 1,024 candidates. Its completeness-certificate identity is `sha256:0530d875a4941751645201917bb60d46a2b0605d5fa734874b2c5f9e8fadfab1`. The table states every generating axis, its unfavorable coordinate and its forced coordinate. Their full row-level Cartesian product is preserved in `claims/SFT-INFO-QUANTUM-CORRESPONDENCE-001/candidate_census.json`.

Structural axis	Rejected alternative	Exact rejection	Forced coordinate
`unit`	`imported-qubit`	A conventional qubit premise would select the carrier.	`one-Fold-distinction` - The minimal Fold's two held labels supply one exact distinction unit.
`support`	`sampled-branches`	Sampling cannot establish complete finite support.	`complete-Fold-word-support` - Every held-label word over every registered position occurs once.
`basis`	`unnamed-state-count`	A count erases which Fold word corresponds to which basis state.	`bijective-word-basis-trace` - Every word has one unique retained basis name and inverse.
`provenance`	`amplitude-only-row`	A coefficient without a generated word erases branch origin.	`position-label-trace` - Every branch retains every canonical position and held label.
`superposition`	`complex-amplitude-premise`	Complex amplitudes, phases and normalization are not derived here.	`complete-alternative-support` - Superposition-equivalence means the exact complete branch family only.
`composition`	`tensor-symbol-premise`	A borrowed symbol does not enumerate joint alternatives.	`complete-pair-cell-support` - Every left branch pairs exactly once with every right branch and retains both origins.
`observation`	`stochastic-collapse`	A random collapse rule is not generated at this boundary.	`exact-observation-class` - The observed image retains its complete branch predecessor class.
`record`	`discarded-predecessors`	Discarding the record prevents exact support reconstruction.	`complete-branch-image-record` - Every pre-observation branch and image remains available for reconstruction.
`generality`	`small-depth-table`	A table does not establish the next support layer.	`position-successor` - A fresh position appends every held label to every prior branch.
`addition`	`extra-amplitude-phase-or-gate-law`	Those structures belong to later forced derivations and would select results here.	`no-extra-quantum-operation` - Only the exact information-support correspondence is admitted.

17.3 Exhaustion, unique survivor and laws

> *The quantum-information correspondence kernel is one Fold distinction per position, complete Fold-word branch support, bijective basis and provenance traces, support-only superposition equivalence, complete pair-cell joint support, exact observation classes, reconstructing records, position induction and no imported quantum operation.*

The engine recorded 1,024 decisions, 1,023 eliminations and exactly one survivor, `one-Fold-distinction__complete-Fold-word-support__bijective-word-basis-trace__position-label-trace__complete-alternative-support__complete-pair-cell-support__exact-observation-class__complete-branch-image-record__position-successor__no-extra-quantum-operation`. Replacing any survivor coordinate with the rejected coordinate triggers the table's failure condition; the extra-rule coordinate prevents an unforced addition. This proves minimality and named-shape uniqueness inside the frozen grammar. The survivor entails:

- one Fold position supplies one independently realized information distinction.
- k positions generate every held-label word exactly once.

- basis correspondence is bijective and retains each Fold word.
- joint support is the complete pair product with both source coordinates retained.
- observation plus the complete branch-image record reconstructs pre-observation support exactly.

17.4 Operational demonstrations and adverse controls

- `complete-branch-support` - Two held labels over three positions generate eight exact unique branches. Result: `PASS`.
- `unit-trace` - Each of three positions independently realizes one Fold information distinction. Result: `PASS`.
- `basis-bijection` - Every branch has one unique retained basis name. Result: `PASS`.
- `support-only-boundary` - Superposition-equivalent support explicitly refuses amplitudes at this boundary. Result: `PASS`.
- `joint-composition` - Two one-position supports generate four exact joint pair cells. Result: `PASS`.
- `record-reconstruction` - First-label observation retains four branches and its record reconstructs all eight. Result: `PASS`.

The engine-level unfavorable controls are:

- `false_premise` - expected: reject a generated form missing the first required structural condition Observed: A conventional qubit premise would select the carrier. Result: `PASS`; receipt `sha256:c421b75768ca85be4d8d1a3b5ed28f6aca315c61acf1d1cc47de8957a467774e`.
- `tampered_source` - expected: reject a changed official source identity Observed: the changed identity differs from the registered source manifest Result: `PASS`; receipt `sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26`.
- `tampered_artifact` - expected: reject a missing, duplicated or additional survivor Observed: the generated product contains exactly one registered survivor Result: `PASS`; receipt `sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219`.
- `boundary` - expected: halt any attempt to import an excluded object or answer-producing model Observed: no complex, irrational or imaginary proof value; no amplitude or phase law; no stochastic collapse postulate; no interference, entanglement or gate semantics claimed at the support boundary Result: `PASS`; receipt `sha256:cd66bc50dc444891797124878cf737140b4cffdea7096e0bbc3446294a36c4bf`.

17.5 Depth-independent closure

Base: The first Fold position has the two held labels, one exact distinction and a two-branch basis correspondence.

Successor: Adding one fresh canonical position appends each held label to every prior word exactly once. Prior branch provenance remains intact, the new position realizes one distinction and joint-support laws remain pairwise.

The executed product closes the registered representation alternatives. The base/successor certificate extends that rule to every generated finite support size, word depth, source extension or action extension named by the claim. It does not convert finite generation into a completed infinity.

17.6 Meaning and scientific consequence

The quantum result is deliberately exact and bounded. One Fold distinction is the native information unit. Complete held-label words across canonical positions correspond bijectively to finite basis support, and complete pair cells supply joint support. 'Superposition-equivalent' means equality of enumerated branch alternatives and provenance only. Observation retains an exact branch class, while the complete branch/image record reconstructs pre-observation support. Amplitude, complex phase, interference, entanglement, gate dynamics and collapse are not claimed here; their absence prevents support correspondence from silently becoming an operational quantum theory.

17.7 Correspondence only after sealing

After the SFT survivor and receipt are fixed, the following established terms may identify translation points: qubit support, computational basis, superposition support, joint quantum state, measurement record. They are not dependencies and did not provide formulas, thresholds or answer tables. A mismatch remains visible as a boundary result rather than being repaired by changing the sealed law.

17.8 Exact exclusions and limitation

- no complex, irrational or imaginary proof value.
- no amplitude or phase law.
- no stochastic collapse postulate.
- no interference, entanglement or gate semantics claimed at the support boundary.

This theorem closes only finite quantum-information support correspondence. Phase, interference, entanglement, measurement dynamics, transformations, gates, circuits and quantum fault tolerance remain quantum-branch obligations.

17.9 Machine evidence identities

- Registered status: `independently_replicated`.
- Source manifest: `sha256:19d64ff5e92da9623652423b85f4d069864f440697b2b6c62e125d987ef7d564`.
- Derivation seal: `sha256:936f0834e7c4b89ea1963b45071b23ab3e5a6aa4097b9f2564d0020746afdf0d`.
- Independent implementation: `sha256:1ca65f3dd9401285c04ed7460d06560ce61479d12dc0d003e0411624e3f0d031`.
- Independent certificate: `sha256:66aa35ac197317adc2f25d7dd83eba7b10710dd192e8f51e5b54270bc5aa8c1b`.
- External validation: `sha256:70288b5837290911752ee6245cc2d6b7ff8c9169859b0adc6403c6e70e6d96da`.
- Engine receipt: `sha256:01279dbd48775d7f5fa92b0c9abe07dcafeb2ba670d10374d9f3bb355582cc81`.
- Engine receipt path: `receipts/engine/model_admitted/SFT-INFO-QUANTUM-CORRESPONDENCE-001-01279dbd48775d7f.json`.

18. Cross-derivation synthesis: one information kernel

The branch is one dependency chain. Canonical symbols and total observation first establish the source alternatives and their distinguishability. Encoding then maps those alternatives to generated words while preserving source classes. Quantity collects complete support and retained/closed pair ledgers. Entropy reorganizes the same evidence by observation class and exact support part. Compression asks whether a complete representation reconstructs the source and counts the entire representation system.

Channels generalize encoding to complete input/output relations and identify confusability by support overlap. Noise adds registered transformation identity and exact changed positions. Coding chooses generated word families whose complete registered error images are separated. Mutual and conditional information then compare two observation partitions through exact joint cells. Conservation classifies every source distinction under transformation, and the classical/probabilistic law shows that one held state and an unresolved observation class use the same carrier.

Quantum-information support correspondence is last because it needs all earlier results: Fold-word support, native distinction units, joint composition, observation classes and conservation records. It cannot retroactively import amplitudes or gates into the symbol law. Likewise, later channel or code correspondences cannot change the exact observation ledger that forced their prerequisites.

19. Entropy without logarithms and probability without randomness

The conventional scalarization of finite information support is not used as foundational evidence. A logarithm needs a chosen base and can produce irrational or floating values outside the admitted proof domain. SFT retains the exact object from which such summaries are ordinarily calculated: every observation class, every member, its exact positive part of the whole and every unresolved pair. A singleton class uses empty One; an absent joint cell uses empty One.

This richer object makes refinement transparent. If one observation refines another, every fine class lies within a coarse class and its closed-pair ledger is a subset of the coarse ledger. Conditional information is exact restriction to a given class. Dependence is the complete comparison between joint cell parts and exact marginal products. No negative deviation is needed: the relation records equal, joint-below-product, joint-above-product or empty-One.

Superdeterminism presents no contradiction here because probability is not a transition cause. The generated microstate support and every registered transformation remain deterministic. Probability describes the part occupied by an observation class when source distinctions are unavailable. A prior, seed or stochastic kernel would be an additional parameter and is rejected unless a later empirical claim separately registers and validates it.

20. Compression, capacity and correction as exact finite support laws

Compression, channel capacity and error correction are often reported by scalar rates. This branch derives the finite objects those rates summarize. A lossless description includes its dictionary, token trace, parser and decoder. A channel includes every registered source/output/path cell. A code includes every registered error image of every codeword. The machine can therefore determine reconstructability, confusability, detection and correction without an error distribution or distance threshold.

The exact capacity object is the complete family of greatest pairwise distinguishable input selections. Retaining the whole family prevents a tie from being hidden by one arbitrarily selected code. The exact correction criterion is disjoint image support; decoding then has one source predecessor. Conventional length, rate, Hamming distance and capacity units may be computed as post-seal correspondence summaries for a declared finite carrier, but they cannot replace the support evidence or select the survivor.

21. Quantum-information boundary and later obligations

The branch closes a precise classical-probabilistic-quantum information boundary. A classical state is one held word. A probabilistic state is an exact observation class over complete deterministic word support. The quantum support correspondence is the complete generated Fold-word alternative family with a bijective basis trace. Pair-cell composition yields complete joint support. Observation and a retained branch/image record yield exact class reduction and support reconstruction.

Support equality does not establish operational equivalence. No complex amplitude, phase rule, interference operation, entangling law, measurement dynamics, unitary transformation, gate, circuit, quantum code or fault threshold is used or claimed. Those are separately enumerated obligations of Reversible and Quantum Computation. This separation is a positive result: it states exactly what information structure is already available and prevents downstream quantum phenomena from being smuggled into the carrier definition.

22. Formal computational evidence versus natural empirical evidence

The claims in this paper are formal structural theorems. Their empirical evidence is computational: complete candidate products are generated, every decision executes, adverse controls are actually perturbed, independent validator processes run and stored receipts are replayable. This validates the reported machine behavior of the declared grammar and certificates. It is not mislabeled as observation of an external natural system.

A later claim about natural measurements must register before target access, seal its formal predictor, enforce target custody, disclose all rows and preserve falsification results. Blind opaque prediction is insufficient because a score does not expose premises, candidate alternatives, eliminated structures, source identity, information leakage or the conditions under which the claim must halt. Application domains may test a sealed law but cannot select or repair it.

23. Complete V1/V2 ownership and same-strength reconstruction

Version 1.1 closes the branch against the complete registered prior record rather than only the twelve V3 headings. The audit read all 356 V1 theorem-manifest rows and all 407 V2 numbered results. Composite source statements were split into atomic obligations so that an information theorem could not absorb an unperformed physical measurement, and a downstream physical interpretation could not make a closed formal information result disappear. Twelve V1 rows and twenty-five V2 steps contain Information Science-owned content. They decompose into 77 atomic obligations; all 77 map to model-admitted V3 receipts at equal strength and none remains open. The authoritative ledger is `census/information_science_prior_obligations.json`.

The reconstruction retains the strongest prior content: one native distinction closes per Fold observation; held labels reconstruct the source; exact equal shares partition the One; entropy is complete observation-class and unresolved-pair structure; predecessor merging is exact information loss; recurrent return retains its distinctions; support uncertainty, branch support, deterministic probability, channel provenance, reverse records, finite coding, conditional reconstruction and the classical-probabilistic-quantum support boundary all remain explicit. None is reduced to a historical note or accepted merely because V1/V2 stated it.

Prior source	Atomic obligations	Admitted V3 law labels	Result
V1 Q5	2	ENCODING-DECODING, QUANTITY	2/2 closed
V1 Q6	1	CLASSICAL-PROBABILISTIC	1/1 closed
V1 PH2	1	ENTROPY-UNCERTAINTY	1/1 closed
V1 D6	1	ENTROPY-UNCERTAINTY, QUANTUM-CORRESPONDENCE	1/1 closed
V1 N7	2	CONSERVATION-LOSS, ENTROPY-UNCERTAINTY	2/2 closed
V1 C2s	1	SYMBOL-DISTINCTION	1/1 closed
V1 C5s	1	SYMBOL-DISTINCTION	1/1 closed
V1 C8s	2	CONSERVATION-LOSS, SYMBOL-DISTINCTION	2/2 closed
V1 I-2	2	ENTROPY-UNCERTAINTY, QUANTITY	2/2 closed
V1 I-10	2	CONSERVATION-LOSS	2/2 closed
V1 G1	2	CLASSICAL-PROBABILISTIC, QUANTUM-CORRESPONDENCE	2/2 closed
V1 G3	2	CHANNEL-CAPACITY, QUANTUM-CORRESPONDENCE	2/2 closed
V2 Step 21	2	CONSERVATION-LOSS, ENTROPY-UNCERTAINTY	2/2 closed
V2 Step 22	1	ENTROPY-UNCERTAINTY, QUANTUM-CORRESPONDENCE	1/1 closed
V2 Step 28	1	CONSERVATION-LOSS	1/1 closed
V2 Step 38	2	CLASSICAL-PROBABILISTIC, QUANTUM-CORRESPONDENCE	2/2 closed
V2 Step 98	2	CONSERVATION-LOSS, ENTROPY-UNCERTAINTY	2/2 closed
V2 Step 123	2	CONSERVATION-LOSS	2/2 closed
V2 Step 159	2	CLASSICAL-PROBABILISTIC, QUANTUM-CORRESPONDENCE	2/2 closed
V2 Step 192	2	CONSERVATION-LOSS	2/2 closed

Prior source	Atomic obligations	Admitted V3 law labels	Result
V2 Step 203	2	CLASSICAL-PROBABILISTIC, ENTROPY-UNCERTAINTY	2/2 closed
V2 Step 252	2	CHANNEL-CAPACITY, QUANTUM-CORRESPONDENCE	2/2 closed
V2 Step 257	1	SYMBOL-DISTINCTION	1/1 closed
V2 Step 326	2	QUANTITY, SYMBOL-DISTINCTION	2/2 closed
V2 Step 328	3	ENCODING-DECODING, SYMBOL-DISTINCTION	3/3 closed
V2 Step 329	5	CLASSICAL-PROBABILISTIC, CONSERVATION-LOSS, ENTROPY-UNCERTAINTY, QUANTITY	5/5 closed
V2 Step 348	2	CLASSICAL-PROBABILISTIC, ENCODING-DECODING	2/2 closed
V2 Step 349	2	CONSERVATION-LOSS, MUTUAL-CONDITIONAL	2/2 closed
V2 Step 356	1	COMPRESSION-REDUNDANCY	1/1 closed
V2 Step 384	3	CLASSICAL-PROBABILISTIC, ENTROPY-UNCERTAINTY	3/3 closed
V2 Step 385	3	COMPRESSION-REDUNDANCY, ENCODING-DECODING	3/3 closed
V2 Step 386	3	CHANNEL-CAPACITY	3/3 closed
V2 Step 387	3	NOISE-ERROR	3/3 closed
V2 Step 388	3	CODING	3/3 closed
V2 Step 389	3	MUTUAL-CONDITIONAL	3/3 closed
V2 Step 390	4	CLASSICAL-PROBABILISTIC, QUANTUM-CORRESPONDENCE	4/4 closed
V2 Step 402	2	CHANNEL-CAPACITY, QUANTITY	2/2 closed

Categorical separation is part of the result. The formal information component of thermodynamic erasure closes here; the physical heat value remains a Physics obligation. Exact branch support and unresolved-part accounting close here; Born dynamics and natural measurement remain Physics and Quantum Computation obligations. Finite support uncertainty closes here; physical position-momentum measurements remain in Physics. Classical held-word copying and quantum-support correspondence close here; operational no-cloning, gates and fault tolerance remain in Quantum Computation. These are retained downstream requirements, not exclusions or deflations.

24. External validation at the correct evidential boundary

Information Science is a formal branch: symbol identity, finite support, observation partitions, code relations and distinction ledgers do not possess a natural measured constant for an external measurement body to supply. Inventing one would be a category error and a free parameter. The applicable external validator is implementation-distinct exact reproduction. For every law, a separate process regenerates the literal grammar, candidate order, decisions, unique survivor, controls and closure certificate without importing the scientific law module; its result must match the sealed receipt exactly or admission halts.

Post-seal comparison to Hartley, Shannon, Hamming, Landauer, Bennett and Schumacher identifies correspondence and difference, not derivational authority [3-8]. The SFT carrier retains complete finite support and exact source provenance; conventional logarithmic quantities, distances and rates can summarize a declared sealed object afterward. Where a later physical claim has units and an authoritative measured value, that owning branch must seal the predictor before target access and compare value, units, uncertainty, source version and every adverse row. Information Science cannot pre-approve that empirical result by analogy.

This distinction is especially important for superdeterminism. An equal support part is not a random force. It reports which exact deterministic alternatives remain unresolved relative to a declared observation. A pseudo-random seed, sampler, Bayesian prior or fitted distribution would be an additional causal or representational input and is rejected unless separately generated and registered. Randomized search can therefore be closed as equal-share accounting over an unresolved deterministic target family while every source-conditioned execution remains fixed.

25. Information power, open knowledge and institutional accountability

Information Science makes the politics of scientific access technically visible. A result that exposes only a score, headline or authority label closes the distinctions needed to test its premises, alternatives, data custody, exclusions and failure conditions. Paywalls, undisclosed training corpora, proprietary validators and inaccessible source data are therefore not merely social inconveniences: they remove information required for independent falsification. An opaque oracle may be useful, but its answer is not a closed theorem unless the evidential route is itself inspectable.

The institutional critique is evidence-based, not a claim that every funded scientist or institution acts in bad faith. Systematic reviews report associations between commercial sponsorship and favorable outcomes and show that sponsorship can shape research agendas [11,12]. Reviews of grant peer review identify limits in the evidence for reliability and fairness [13,14]. The scientific record has also underrepresented null and negative results [15]. These mechanisms show how capital, prestige, publication incentives and access control can become misaligned with complete public evidence. Consensus and expertise remain valuable sources of criticism; neither is an admission certificate.

Ernos Labs answers that problem by attaching scientific authority to inspectable artifacts and revocable conformance. The paper, ledgers and evidence maps are openly readable under CC BY 4.0; executable code is Apache-2.0; the Ernos Labs designation is available only while a work obeys the public empirical constitution. Copyright preserves Maria Smith's authorship and prevents appropriation from being confused with openness; it does not prohibit reading, redistribution, criticism, reproduction or derivative scholarship under the stated licences. UNESCO's open-science recommendation likewise treats scientific knowledge, infrastructures, communication and engagement as public-facing concerns [16].

Maria Smith developed this programme without conventional academic credentials, institutional appointment or a funding gate granting permission to speak. That biography is not evidence that any theorem is true; the public derivations and receipts must carry that burden. It is evidence of what credential-first filtering risks excluding. The point is not a celebration of one outsider but an indictment of every mind lost when access, status or capital is treated as a proxy for the ability to produce testable knowledge. Independent researchers, academic specialists and institutional reviewers are invited to reproduce, invalidate and improve the work on equal evidential terms.

26. Machine reproduction and current verification

The definitive cross-platform logical command is:

```
python3 -m sft verify-all   # macOS and most Linux systems
py -m sft verify-all        # standard Windows Python launcher
```

No Docker image, virtual machine, network service or third-party Python package is required. The verifier validates repository structure, runs unit, unfavorable-control and end-to-end tests with 100 percent executable-line coverage of the core admission engine, rebuilds source manifests, replays the dependency-ordered census, launches every implementation-distinct validator and compares each result with its stored receipt.

The branch-specific release verification for this version is:

```
SFT INFORMATION SCIENCE PUBLICATION GATE: PASS
live claims: 12
frozen inventory claims: 12
paper-covered claims: 12
generated candidate structures: 11,776
V1/V2 atomic Information Science obligations: 77/77 closed
```

The repository-wide 100 percent engine-coverage baseline remains an invariant of the platform, but it was not rerun for inconsequential prose changes in this patch. The minimal release gate rechecked branch inventory equality, model-admitted receipts, complete evidence-map coverage and paper hashes. Scientific closure additionally requires the claim-specific grammar, exhaustive decision vector, unique survivor, minimality, named-shape uniqueness and induction certificate. Neither test coverage nor a cryptographic hash substitutes for scrutiny of the registered scientific boundary.

27. Reproduction, criticism and invalidation routes

A reviewer can challenge completeness by producing a structurally possible coordinate inside the stated boundary that the registered domains omit. Minimality fails if a rejected coordinate preserves every registered requirement. Uniqueness fails if a second generated member survives. Depth-independent closure fails if the base or successor certificate has an internal counterexample. An operational law fails if a valid input inside the boundary violates it.

Mechanical invalidation is equally direct: change an official source, alter candidate order, remove a decision, add a survivor, fail a control, reorder dependencies, tamper with a certificate or change a receipt. The engine halts at the violated gate. Hashes preserve artifact identity; they do not establish scientific truth. The evidence map makes it possible to distinguish disagreement with an exact artifact from disagreement with the grammar itself.

28. Scope, limitations and next branch

Closure is exact at the frozen generated-finite Information Science boundary. The paper does not claim a completed infinite source, every asymptotic coding theorem, every empirical communication medium or a completed operational quantum theory. Named applications must still register their supports, transformations, resources and empirical protocols. Conventional scalar summaries may be calculated only as explicit post-seal correspondences.

The next dependency branch is Formal Computation: state and transition, formal languages and grammars, automata, rewriting systems, recursive functions, lambda-like calculi, abstract machines, circuits, operational processes, composition and decomposition, equivalence among computational models and universality. Quantum operations remain reserved for the later Reversible and Quantum Computation branch.

29. Open-science status, rights and participation

The platform is openly inspectable, reusable and redistributable under its published licences. Maria Smith retains copyright and authorship. Paper and documentation text is CC BY 4.0; code is Apache-2.0. The Ernos Labs designation is a separate conformance mark requiring adherence to the scientific constitution, transparent evidence, fail-closed engine rules and community conduct policy.

Credentials cannot rescue a failed gate, and lack of credentials cannot prevent a reproducible counterexample from being evaluated. Independent replications, omitted candidates, counterexamples and submissions are invited through Maria.Smith.Sftoe@gmail.com and https://discord.gg/ucwGryVxGr. Maria Smith explicitly authorized this same-paper successor release on 24 July 2026.

30. Conclusion

The Information Science branch closes twelve dependency-ordered kernels through 11,776 generated candidates, twelve unique survivors, twelve depth-independent certificates and twelve implementation-distinct validations. Its complete prior audit reviews 763 V1/V2 source entries and closes all 77 Information Science-owned atomic obligations at equal strength. It reconstructs the progression from symbols and observation to representation, quantity, uncertainty, compression, channels, error, coding, dependence, conservation and classical-probabilistic-quantum support correspondence without importing conventional answer-producing models.

Its unifying result is exact distinction accounting. Every alternative has a generated source, every observation declares what it retains and closes, every transformation accounts for its predecessor fibres and every correction must have a unique source. The branch is admitted within that exact boundary because its complete machine-auditable chain closes - not because a prior authority, familiar scalar or application score selects it.

Appendix A. Authoritative receipt identities

Claim	Engine receipt
SFT-INFO-SYMBOL-DISTINCTION-001	sha256:e387d64c93f5241d90ecf2a447eb6a3518c1e9f473e7ec627d38b7f03ac9d941
SFT-INFO-ENCODING-DECODING-001	sha256:dc3cbc708f5b88b3d9bbae5a646a644dc8feb7fe4a5d6f8367052f5062846d0f
SFT-INFO-QUANTITY-001	sha256:a06381112b197ea1762d8dd06197c87f03055a4554429f100ead4a93bbb2362d
SFT-INFO-ENTROPY-UNCERTAINTY-001	sha256:d604932214499ea1e11fc2df7b71f4ea0b67014e3001023720b325d6203d2b72
SFT-INFO-COMPRESSION-REDUNDANCY-001	sha256:4d46ed451a5cd3c3a19fcb9d8b8cc3c32ffc8f92a9cc1d55cfb93f331023ee07
SFT-INFO-CHANNEL-CAPACITY-001	sha256:fb47c7f7be3e832cb67658042b2c7c3224113d9262cb4c45df7924a1f3a1cc01
SFT-INFO-NOISE-ERROR-001	sha256:bc42a3ab0291a59b278654bed369bb1d64b6abe1f9278af9f89eac78211f48b9
SFT-INFO-CODING-001	sha256:7c7648bdba686025c8b1f647b3e63a242da24234aa953cac184021f878b07711
SFT-INFO-MUTUAL-CONDITIONAL-001	sha256:c311f27de12d70ea2307acb2d5b23f83cb065b6b28fe5c08ac4742f6f1aadc26
SFT-INFO-CONSERVATION-LOSS-001	sha256:4e2892e85c1624a2234163fdd51377a7086219ce0945cbf6acfcaa41664303f4
SFT-INFO-CLASSICAL-PROBABILISTIC-001	sha256:ce0515fdf69d368bddb313faad3a88043550e90e3df21c4049267e1a5e54fe7c
SFT-INFO-QUANTUM-CORRESPONDENCE-001	sha256:01279dbd48775d7f5fa92b0c9abe07dcafeb2ba670d10374d9f3bb355582cc81

Appendix B. Complete derivation and evidence route

For every claim identifier `<CLAIM>`, the complete route is:

```
claims/<CLAIM>/registration.json
claims/<CLAIM>/WHY_DERIVATION_CHECK.md
claims/<CLAIM>/candidate_census.json
claims/<CLAIM>/elimination_receipt.json
claims/<CLAIM>/controls.json
claims/<CLAIM>/certificate.json
claims/<CLAIM>/execution.py
claims/<CLAIM>/independent_validator.py
receipts/engine/model_admitted/<CLAIM>-<receipt-prefix>.json
```

Executable law sources are under `sft/information_science/`; the catalog fixes dependency order; the model census is `census/claims.json`; the replay order is `census/execution_manifest.json`; and the frozen scope is `publications/inventories/information_science.json`. The paper evidence map binds every derivation section to exact claim-package and receipt hashes.

Appendix C. Reproducibility interpretation

A successful replay establishes that the checked sources reproduce the registered candidate products, decisions, closures, controls, independent certificates and receipts on the reviewing host. Scientific review must also ask whether the frozen grammar exhausts its stated boundary. Reproduction and boundary criticism are complementary, and the repository preserves evidence for both.

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